



StreamCaster Programming Manual

Version 4.3
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Revision History

Version	Date	Changes
0.1	March 20, 2013	Initial draft
0.2	March 25, 2013	Added RS232 config, antenna mask, auto ranging, replaced command table with paragraphs, etc.
0.3	March 28, 2013	Added appendices for MCS tables, an example RSSI report, network status command, and other minor changes.
0.4	April 16, 2013	Removed Virtual IP commands, added antenna selection, transmission disable, Qos and other minor changes.
0.5	May 9, 2013	Updated descriptions on serial configuration, network status. Added radio reset, radio mode as well as commands related to continuous waveform and spectrum scanning.
0.6	May 18, 2013	Added RX antenna option for spectrum scanning and clarified parameters.
0.7	May 24, 2013	Added age parameter to Time of Flight Command, updated description.
0.8	July 8, 2013	Added GPS Coordinates and Burst Time command.
0.82	August 8, 2013	Fixed -f format on Spectrum Scan
0.9	August 23, 2013	Added commands related to Virtual IP and external PA. Updated Spectrum Scan Stop command
0.91	October 10, 2013	Added -g option for spectrum scanning command
0.92	December 11, 2013	Updated external pa support API.
0.93	December 19, 2013	Updated routing beacon period, supported frequencies API
0.94	January 17, 2014	Updated supported frequencies command
0.95	March 10, 2014	Several wording fixes
0.96	March 14, 2014	Added supported firmware version and additional information on savenvlinsingle command
0.97	March 17, 2014	Changed supported firmware version
0.98	May 8, 2014	Changed supported firmware version
1.0	December 9, 2014	Added support for uploading custom logo/manual
1.1	August 26, 2015	New command and API access methods
1.2	October 15, 2015	Removed unsupported commands

1.3	November 11, 2015	Added neighbor information and encryption profile commands
1.4	December 4, 2015	Added instructions for using curl to access API on Windows™, discontinued supported frequencies table
1.5	December 30, 2015	Removed unsupported commands, added command for PA max output per channel
1.6	April 18, 2016	Removed unsupported commands and modified default values for burst time
1.7	May 23, 2016	Added 10MHz to list of officially supported bandwidth
1.8	December 29, 2016	Added noise level, neighbor RSSI, routing path and FPGA build info API
1.9	January 11, 2017	Added password protected API section
2.0	February 14, 2017	Added enable max power API
2.1	April 16, 2018	Added FIPS mode supporting APIs
2.2	May 10, 2018	Added PTT APIs
2.3	June 13, 2018	Added QoS Scheduler APIs
2.4	September 17, 2018	Added Multicast APIs
2.5	December 13, 2018	Fixed typos and confusing text
2.6	March 14, 2019	Added throughput APIs
2.7	April 19, 2019	Added GI Mode APIs
2.8	June 24, 2019	Added MAN-IA and other missing APIs
4.0	October 13, 2020	Changed naming to be consistent with SS4 firmware
4.1	November 22, 2021	Added SS4 APIs. Sections 3.122 through 3.161 and Section 10.
4.2	January 17, 2022	Added API for SSv4.0.3.0. Sections 3.165-3.178
4.3	October 3, 2022	Update manual in FIPS effort
4.3.1	March 31, 2023	Added fips_ready API. Added Section 12 - API User Privilege
4.3.2	May 3, 2023	Added deferred_execution_api API. Noted limitations for freq and bw APIs.
4.3.3	May 5, 2023	Added Section 12.3 No Restriction APIs. Added Read Output Format to all APIs
4.3.4	Jul 18, 2023	Added APIs: routing_tree, qos_class_cwmin, qos_class_cwmax
4.3.5	Aug 1, 2023	Added API: bcast_discovery_config
4.3.6	Aug 22, 2023	Shortened duplicated 3.64 section.

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1 Objective

The objective of this document is to provide information about the StreamCaster radio API.

2 Programming Interface

The StreamCaster radio exposes the API via an HTTP web service running on the radio:

- Users can access this API on port 80 by invoking HTTP POST on the CGI command "streamscape_api".
- The requests and responses are encoded in JSON-RPC version 2.
- As specified in JSON-RPC v.2, each request object will contain (among other members) a string "method" and an optional object "params".
- "method" indicates the specific command being sent to the radio and "params" contains optional parameters encoded as a JSON array of strings.
- The response object from the radio will contain a "result" and/or an "error" field, "result" is a JSON array while "error" is a JSON object.
- If the command is successful, the response object will set "result", otherwise "error" will be set to an appropriate value.

2.1 HTTP access

- Bring up the radio with IP address 172.20.XX.XX and connect a laptop to the ethernet interface.
- Access port 80 using curl to access the API directly using commands.
- Below are a few example commands/outputs, any commands should be in a single line:

Read Current Frequency:

Command:

```
$ curl -i -X POST -d '{"jsonrpc":"2.0","method":"freq","id":"sbkb5u0c"}' http://172.20.XX.XX/streamscape\_api
```

Output:

```
HTTP/1.1 200 OK
Content-Length: 115
Content-Type: application/json
{
  "result" : ["2492"],
  "id" : "sbkb5u0c",
  "jsonrpc" : "2.0"
}
```

Set RSSI Report Address:

Command:

```
$ curl -i -X POST -d '{"jsonrpc": "2.0", "method": "rssi_report_address", "params": ["192.168.10.100", "2000"], "id": "sbkb5u0c"}' http://172.20.XX.XX/streamscape\_api
```

Output:

```
HTTP/1.1 200 OK
Content-Length: 115
Content-Type: application/json
{
  "result": "",
  "id": "sbkb5u0c",
  "jsonrpc": "2.0"
}
```

Note: If using curl on Windows™, the following changes are needed:

- The entire JSON bracket {} need to be enclosed with double quotes "", instead of single quotes "
- All the double quotes in JSON bracket {} need to be changed from " to \"

An example is given below:

Input:

```
C:\>curl -i -X POST -d '{"jsonrpc":"2.0","method":"freq","id":"sbkb5u0c"}' http://172.20.XX.XX/streamscape\_api
```

Output:

```
HTTP/1.1 200 OK
Access-Control-Allow-Origin: *
Content-Type: application/json
Transfer-Encoding: chunked
Date: Sun, 02 Jan 2000 18:46:01 GMT
Server: lighttpd/1.4.32
```

```
{"result": ["4920"],"id": "sbkb5u0c", "jsonrpc": "2.0"}
```

2.2 Serial/SSH API access

API access is also allowed via:

- Serial: Serial interface needs to be programmed into “Debug” mode. The serial console is available at 115200 baud rate, 8N1 settings
- SSH: Login to the radio using an SSH client with username and password root. This also works for remote radios.

Upon login via serial/SSH the command prompt shows:

(Cmd)

At this prompt, user may type:

```
api IP CMD_NAME 'COMMAND_PARAMETERS'
```

IP: IP of radio you want to run API on. If local radio, use 127.0.0.1 or radio IP address

CMD_NAME: API command name

PARAMETERS: JSON style parameters enclosed by single quotes '

Below is an example for setting serial settings:

```
### read back serial settings
(cmd) api 127.0.0.1 serial_config
["115200", "8", "2", "0", "0", "0", "172.20.4.223", "udp", "54321"]
### modify serial settings (changed peer ip address)
(cmd) api 127.0.0.1 serial_config ["115200", "8", "2", "0", "0", "0", "172.20.4.111", "udp", "54321"]
[""]
### read back serial settings
(cmd) api 127.0.0.1 serial_config
["115200", "8", "2", "0", "0", "0", "172.20.4.111", "udp", "54321"]
```

3 Commands and Responses

Unless otherwise specified, all commands set/reads a certain variable. As stated above, commands should be filled in the "method" field. If no arguments are given, "result" will be set to the current variable value as a JSON array. If "params" is set to a certain value, the radio will change the corresponding variable to the given value. If the change is successful, result is set to [""] (JSON array containing an empty string). Otherwise "error" will be set to a string message indicating the failure.

Below is a list for the available commands:

3.1 Node ID

- Command: nodeid
- Parameter: N/A, since this variable is read-only.
- Description: Every radio has a unique 16-bit node id. This command reads the node-id of the radio. This id cannot be changed by the user.
- Read Output format: ["12345"], integer value in quotes

3.2 Frequency

- Command: freq
- Parameter: ["frequency"], where frequency is a number.
- Description: Sets/reads the current frequency of the radio.
- Unit: MHz
- Range: Use Supported Frequency command
- Resolution: 0.1
- Read Output format: ["2490"], the value in quotes is the current frequency in floating point.
- Note: This will cause a soft boot. The driver and services will reboot and become unavailable. Please refrain from issuing other API call during the execution of this API.

3.3 Bandwidth

- Command: bw
- Parameter: ["bandwidth"], where bandwidth is a number.
- Description: Sets/reads the current bandwidth of the radio.
- Unit: MHz
- Range: 5 or 10 or 20 (IVAS radios range: 1.25, 2.5, and 5 MHz)
- Default: 20MHz (IVAS radios default: 5 MHz)
- Read Output format: ["20"], the value in quotes is the current bw in floating point.
- Note: This will cause a soft boot. The driver and services will reboot and become unavailable. Please refrain from issuing other API call during the execution of this API.

3.4 Transmit Power

- Command: power_dBm
- Parameter: ["power"], where power is an integer number.
- Description: Sets/reads the current transmit power of the radio. If "enable_max_power" is 1, the value of power_dBm is ignored.
- Unit: dBm
- Range: 0-39, note however that the actual power output is accurate only between 10-39 dBm. The upper limits of power may vary depending on radio model and installed license.
- Resolution: 1
- Default: 30 dBm
- Read Output format: ["10"], integer value in quotes

3.5 Max Link Distance

- Command: max_link_distance
- Parameter: ["distance"], where distance is an integer.
- Description: Sets/reads the max link distance of the radio, should be in sync with all radios in the network.
- Unit: meter
- Range: 0~1000000
- Resolution: 1
- Default: 1000 meters
- Read Output format: ["1000"], integer value in quotes

3.6 Encryption

- Command: enc_disable
- Parameter: ["disable"], where disable is either 0 or 1.
- Description: Enables/disables the AES encryption, 1 for disable and 0 for enable, should be in sync with all the radios in the network. When the encryption is disabled, data will be transmitted in plain text.

- **Note: Starting with v3.15.1, after calling `enc_disable`, the API command `enc_disable_confirm` also needs to be called with the same parameter above. This will finalize the encryption setting. Until this API is called, the radio will not switch encryption modes. Please see the description for `enc_disable` in Section 3.83. Until the `enc_disable_confirm` is called correctly, the `enc_disable` API will read-back the previous mode.**
- E.g. current mode is `enc_disable = 0` (Enabled). To disable encryption, first call API command "`enc_disable 1`", then call API command "`enc_disable_confirm 1`". If `enc_disable` is read back before `enc_disable_confirm` is called, it will still return "0".
- Default: 1
- This API is readable and writeable.
- Read Output format: ["1"], integer value in quotes

3.7 Encryption Confirm

- Command: `enc_disable_confirm`
- Parameter: ["disable"], where disable is either 0 or 1.
- Description: Confirms the enables/disable of the AES encryption, 1 for disable and 0 for enable. This command needs to be called after calling `enc_disable`, with the same parameter that was passed before to that call. If the parameter does not match, the command will return an error. Once the command is executed, the radio will switch encryption mode.
- This is a write-only parameter.

3.8 Encryption Key

- Command: `enc_key`
- Parameter: ["enc_key"], where `enc_key` is a string.
- Description: Sets/reads the AES encryption key, should be in sync with all the radios in the network.
- Range: Alpha-numeric string, basically anything available on the keyboard. Length of 64, 128, or 256 depending on the encryption profile.
- Default: "HelloWorld"
- Read Output format: ["HelloWorld"], string value in quotes

3.9 Encryption Key Length

- Command: `enc_key_len`
- Parameter: ["len "], where len is a number.
- Description: Sets/reads the AES encryption key length in bits, should be in sync with all the radios in the network.
- Unit: bit
- Range: 64, 128 or 256 depending on encryption profile.
- Default: 128-bit.
- Read Output format: ["128"], integer value in quotes

3.10 RSSI Report

- Command: rssi_report_enable
- Parameter: ["enable"], where enable is either 0 or 1.
- Description: Enables/disables the RSSI report feature, 1 for enable and 0 for disable.
- Default: 0
- Read Output format: ["0"], integer value in quotes

3.11 RSSI Report Address

- Command: rssi_report_address
- Parameter: ["IP","Port"], where IP is an IP address in the format of XXX.XXX.XXX.XXX, port is an integer.
- Description: Sets/reads the RSSI report IP address and port.
- Range: XXX in the IP address should be between 0 and 255, port number cannot exceed 65535, it is suggested to use a port number around 2000.
- Default: "10.1.1.4", "30000"
- Read Output format: ["10.1.1.4", "30000"], string and integer values in quotes

3.12 RSSI Report Period

- Command: rssi_report_period
- Parameter: ["period"], where period is an integer.
- Description: Sets/reads the RSSI report time period.
- Unit: millisecond.
- Range: 10~1000.
- Resolution: 1
- Default: 10ms
- Read Output format: ["10"], integer value in quotes

3.13 RTS

- Command: rts_disable
- Parameter: ["disable"], where disable is either 0 or 1.
- Description: Enables/disables the RTS/CTS, 1 for disable and 0 for enable. This command is now deprecated, hence disabling RTS/CTS will cause throughput degradation. API no longer available.
- Default: 0
- Read Output format: ["0"], integer value in quotes

3.14 MCS

- Command: mcs
- Parameter: ["mcs"], where mcs is an integer.
- Description: Sets/reads the MIMO modulation and coding rate. Setting it to 255 will enable rate adaptation.

- Range: see Table 5 in Appendix for the supported MCS. 0-15 or 255.
- Default: 255
- Read Output format: ["255"], integer value in quotes

3.15 Neighbor MCS

- Command: nbr_mcs
- Parameter: ["nodeid"], where nodeid is an integer.
- Description: Reads the MIMO MCS used by the link adaptation algorithm to transmit to a node with a direct wireless link to the radio. Must supply a node id as a parameter. Do not use this command if the radios are not running link adaptation algorithm (auto MCS).
- Range: 0-65535
- This API is read-only.
- Read Output format: ["12"], integer value in quotes

3.16 Wired Backbone Gateway

- Command: wbg_disable
- Parameter: ["state"], where state is either 0 (enable), 1 (disable), or 2 (auto).
- Description: This command enables/disables a radio to be a wired backbone gateway. When enabled, StreamCaster radios act as network switches, forwarding all packets to and from the Ethernet to the wireless interface. Hence connecting more than one radio to the same switched network can lead to broadcast storms. Disabling this prevents a radio from operating as a network switch but will retain its packet routing capabilities.
- When set to Auto, the radio will inspect neighboring nodes with a wired backbone connection. If the radio detects a neighbor has enabled its gateway, it will set itself to disable. Else it will find the neighbor with the lowest nodeid, the radio with the lowest nodeid will enable its backbone connection, the rest will disable their backbone connection. The purpose of this logic is to avoid loops in the routing.
- Default: 2 (auto)
- Read Output format: ["2"], integer value in quotes

3.17 Temperature Reporting Address

- Command: temp_reporting_address
- Parameter: ["IP","Port"], where IP is an IP address in the format of XXX.XXX.XXX.XXX, port is an integer.
- Description: Sets/reads the temperature reporting IP address and port.
- Range: XXX in the IP address should be between 0 and 255, port number cannot exceed 65535, it is suggested to use a port number around 2000.
- Default: "10.1.1.3", "30000"

- Read Output Format: ["10.1.1.3", "30000"], string and integer values in quotes

3.18 Temperature Reporting Threshold

- Command: temp_reporting_min_threshold and temp_reporting_max_threshold
- Parameter: ["temp"], where temp is an integer.
- Description: Sets/reads the Min/Max temperature threshold. When the temperature inside the radio is between these two thresholds it is considered to be in "heating" state. When the temperature exceeds the "temp_reporting_max_threshold" it is considered to be in "overheating" state.

When the radio is in

- Range: It is suggested to set "temp_reporting_max_threshold" between 85~90, and "temp_reporting_min_threshold" around 70. Valid range 0-110.
- Unit: Degree in Celsius.
- Default: min 75, max 85
- Resolution: 1
- Read Output format: ["75"], integer value in quotes

3.19 Temperature Reporting Period

- Command: temp_reporting_period
- Parameter: ["period"], where period is an integer.
- Description: Sets/reads the temperature reporting time period. Reports are sent to temp_reporting_address every "period" seconds.
- Unit: second.
- Range: 1-600 (upper limit of 32bit integer)
- Resolution: 1
- Default: 5 seconds.
- Read Output format: ["5"], integer value in quotes

3.20 Temperature Reporting Mode

- Command: temp_reporting_mode
- Parameter: ["mode"], where mode is either 0, 1, or 2.
- Description: Sets/reads the temperature report mode. 0 disable the report, 1 enables the report only when the radio is heating or overheating, 2 sends the report periodically every temp_reporting_period seconds.
- Default: 0 (disable)
- Read Output format: ["0"], integer value in quotes

3.21 Read Current Temperature

- Command: read_current_temperature
- Parameter: N/A, since this variable is read-only.
- Description: Reads the current temperature of the radio.

- Unit: Degree in Celsius.
- Resolution: 1
- Read Output format: ["29"], integer value in quotes

3.22 Maximum Speed

- Command: max_speed
- Parameter: ["speed"], where speed is an integer.
- Description: Sets/reads the maximum speed variable, this setting will depend on channel environment. This setting optimizes the system in the existence of Doppler spread. It is generally only applicable during ground to ground mobile application.
- Unit: miles per hour.
- Range: 0-1000
- Default: 0 mph
- Read Output format: ["0"], integer value in quotes

3.23 Fragmentation threshold

- Command: aggr_thresh
- Parameter: ["thresh"], where thresh is an integer.
- Description: Sets/reads the fragmentation threshold. An incoming packet of the size larger than the specified threshold is fragmented.
- Unit: byte
- Range: 200, 400, 800, 1600
- Default: 1600 bytes
- Read Output format: ["1600"], integer value in quotes

3.24 Current Time of Flight

- Command: current_tof
- Parameter: N/A, since this is a read command.
- Description: Reads the time of flight (distance in time) to all available nodes with direct wireless connection to the radio. "result" is set to a JSON array in the following format: ["nodeid1", "tof1", "age1", "nodeid2", "tof2", "age2" ...]. nodeid1, nodeid2, etc. are replaced by the respective node ids of the nodes and tof1, tof2, are the time of flight to that node. age1 and age2 indicated how old the report is, basically the time between the generation of the report and the time that the users uses the API to fetch the report.

Time of flight feature is designed such that it reports the last known time of flight value of each node:

- This means if 2 radios are connected and had traffic (meaning generated valid time of flight reports), when they disconnect without a reset, the API command will still report the last known time of flight value.
- On the other hand, if 2 radios are connected (have a link, as reflected by the "network_status" command) but never had traffic, there will still be time of flight values reported but the value is zero.

- If 2 radios never have a link, the time of flight report is a JSON array with an empty string in it.

- Unit: 100 nano seconds for time of flight values, 1 mili second for age value.
- Resolution: 1 micro second for time of flight values.
- Read Output format: ["22102","0","588916550","35689","0","589151130"], integer values in quotes. Each trio of values corresponds to a different node.

3.25 Save Variables to FLASH

- Command: setenvlinsingle
- Parameter: ["command"], where command is any of the commands mentioned in this section excluding the read only commands.
- Write only API.
- Description: Saves the values entered using any of the commands in this section into FLASH memory. Reads whatever value currently in the file system and writes it into permanent storage, the actual values cannot be changed with this command. After a reboot of the radio, variables stored using this command will be loaded as the default values.

This command is compatible with all writeable APIs.

3.26 Version

- Command: version
- Parameter: N/A, since this variable is read-only.
- Description: Returns the version of the API as a float.
- Read Output format: ["0.115"], floating point value in quotes

3.27 Quality of Service

- Command: qos_class
- Parameter: ["priority_number", "protocol", "port_range or port_number"], where priority number is an integer between 0 and 10, protocol is 0 or 1 or 2, port_range is of the format XX-XX, where XX are port numbers in integers, port_number is simply an integer. This command supports appending several port numbers. For example, the following parameter will set port 1000-2000, 2010, 2020 to priority 5:
 - ["5", "0", "1000-2000", "2010", "2020"]
- Description: Sets/reads the priority number for each port. There are 11 priorities available, with 0 being the lowest and 10 being the highest, by default each port has priority 0. Setting one port on priority 10 and another on 0 will introduce much more priority difference than setting them on 10 and 9, the user has the ability to fine tune the priority difference by trying different priorities.

Protocol can be set to 0, 1 or 2, which represents UDP, TCP, or both. To read the current Qos setting, the user need to supply a priority number as well as a protocol number as the parameter, the system will return the ports using that priority/protocol in the format of a JSON array like the following:

- ["1000-2000", "2010", "2020"]

If the user enters 2 commands with the same priority and same protocol number, the second command will completely overwrite the first command, any effect of the first command will be undone.

- Read Output format: ["1000-2000", "2010", "2020"], integer list of ports

3.28 Quality of Service Reset

- Command: qos_class_reset
- Parameter: ["priority_number", "protocol"], where priority number is an integer between 0 and 10, protocol is 0 or 1 or 2. Not all parameters are necessarily, for detail reason see the description below.
- Description: Resets the priority number to default (0). By default each port has priority 0 on both udp and tcp. User may use the command gos_class to assign a different priority number for a certain port. The design of this command is meant to remove the changes made by the qos_class command, and can be used in the following ways:
 - If this command is executed without a parameter, all changes made by previous execution of the qos_class command will be cleared. Every port will thus have priority 0 on both udp and tcp.
 - If this command is executed with a priority number supplied as the parameter, it will remove only settings related to this priority number
 - If this command is executed with a priority number as well as a protocol number supplied as parameters, only settings related to this priority and protocol will be removed.
- Write only command.

3.29 Transmitter Antenna Selection

- Command: tx_ant_mask
- Parameter: ["mask"], where mask is an integer.
- Description: Sets/reads the on/off of individual antennas. This number when converted to binary will be a 4 bit-number, where each bit determines the on/off of an antenna. For example, 15 (1111 in binary) will turn on all 4 antennas, while 3(0011 in binary) will turn on antenna 1 and 2 only.
- Range: 0~15
- Resolution: 1
- Read Output format: ["15"], integer value in quotes

3.30 Receiver Antenna Selection

- Command: rx_ant_mask
- Parameter: ["mask"], where mask is an integer.

- Description: Sets/reads the on/off of individual antennas. This number when converted to binary will be a 4 bit-number, where each bit determines the on/off of an antenna. For example, 15 (1111 in binary) will turn on all 4 antennas, while 3(0011 in binary) will turn on antenna 1 and 2 only.
- Range: 0~15
- Resolution: 1
- Read Output format: ["15"], integer value in quotes

3.31 Serial Port Configuration

- Command: serial_config
- Parameter: ["baud_rate", "data_bits", "parity", "stop_bits", "software_flow_control", "hardware_flow_control", "IP", "protocol", "port"], where software_flow_control, hardware_flow_control, and protocol are either tcp or udp, IP is of the format XXX.XXX.XXX.XXX, the rest are integer numbers.
- Description: Sets/reads the current serial port settings, including Baud Rate, Data Bits, Parity, Stop Bits, Software Flow Control, Hardware Flow Control, IP, Transport Protocol, Port. If no parameters are given, the current configurations will be returned in a JSON array, following the order mentioned above. If no priority is set for the port number assigned to the serial port, RS232 will have priority 5 like all the other ports. If the user requires a higher/lower priority for serial port data, the Qos command described in section 3.27 should be used to choose a smaller/larger number for the port number defined here.
 - Beware that even though Stop Bits is either 1 or 2, it's actually represented by 0 and 1 in the API. If you want to configure Stop Bits to 1, you have to enter 0 in the API. On the other hand, when using API to read the configuration, if 0 is returned, it means Stop Bits is 1.
- Range:
 - Baud Rate is either 110, 300, 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200 bps.
 - Data Bits is between 5~8.
 - Parity is 0 for Odd, 1 for Even, 2 for None.
 - Stop Bits is either 1 or 2. (enter 0 or 1 when using API)
 - Software Flow Control and Hardware Flow control are 1 for enable, 0 for disable.
 - Peer IP is of the format XXX.XXX.XXX.XXX, where XXX are integers between 0 and 255.
 - Transport Protocol is tcp for TCP and udp for UDP
 - Peer Port is an integer between 0~65535.
- Default:
 - Baud Rate is 115200.
 - Data Bits is 8.
 - Parity is 2 (None).
 - Stop Bits is 1. (represented as 0 in API)
 - Software Flow Control and Hardware Flow control are 0 (disabled).
 - IP is 0.

- Transport Protocol is tcp.
- Port is 54321.
- Read Output format:
["115200","8","2","0","0","0","172.20.255.110","udp","54321"], list of integers, and string for ip and transport protocol

3.32 Network Status

- Command: network_status
- Parameter: N/A, since this variable is read-only.
- Description: Each radio maintains the complete list of all wireless links in the network. The output JSON array of this command will be something like:

```
["X","Y","48.000000","X","Z","57.000000"]
```

Here X, Y and Z are the node-ids of the radios. The node-id of a radio can be obtained by combining the last 2 bytes of the IP address of the radio into a 16-bit number, e.g. IP address 172.20.4.31 translates to node-id 1055 ($4 \times 256 + 31$). The number after "label" is the SNR (in units of dB) of the wireless link between the two nodes.

- Read Output format: ["22103","41238","34","22103","22108","22"], list of integer node id pairs with their corresponding SNR as a floating point. Each trio of values are linked.

3.33 TX FIFO

- Command: tx_fifo_disable
- Parameter: ["disable"], where disable is either 0 or 1.
- Description: This command enables/disables the radio's ability to initialize a transmission. If set to 1, the MAC driver will prevent anything from going into the transmitter FIFO, thus preventing any transmission to be initialized. Setting it to 0 will restore the transmitting ability. Using this command in combination with the command described in section 3.34 can prevent any packet going out from the radio.
- Default: 0
- Read Output format: ["0"], integer value in quotes

3.34 Response

- Command: response_disable
- Parameter: ["disable"], where disable is either 0 or 1.
- Description: This command enables/disables the radio's ACK/CTS response. If set to 1, the radio will not send any ACK/CTS packet, while setting it to 0 will restore the ACK/CTS ability. Using this command in combination with the command described in section 3.33 can prevent any packet going out from the radio. The value of this command cannot be saved to FLASH.
- Default: 0

- Read Output format: ["0"], integer value in quotes

3.35 Radio Reset

- Command: radio_reset
- Parameter: ["reboot_delay"], where reboot_delay is an integer in seconds
- Description: This command reboots the radio. The user can supply a delay time in seconds as the input parameter. The API will return if the reboot command is accepted correctly, and will perform the reboot after the delay time chosen by the user. If no parameter is supplied by the user, reboot will be performed immediately without returning.
- Range: 0-10
- Write-only.

3.36 Radio Mode

- Command: radio_mode
- Parameter: ["mode"], where mode is either 0 or 1.
- Description: This command sets the radio mode. If set to 0, the radio operate in network mode, while setting it to 1 puts the radio in PHY test mode. PHY test mode is used solely for testing and verifying PHY features, while network mode is the default operating mode supporting networking under mesh environment.
- Default: 0
- Read Output format: ["0"], integer value in quotes

3.37 Continuous Wave Start

- **Command Deprecated**
- Command: cw_start
- Parameter: ["arguments"], where arguments is a string with several parameters.
- Description: This command will enable the radio to send a continuous wave until stopped by the "Continuous Wave Stop" command. Before using this command, the radio should be put in PHY test mode, otherwise it will return an error. The arguments are explained below with an example in the end:

-f -f <freq> sets center frequency, in MHz.

-g sets tx VGA gain.

-g <vga> to set a common gain for all TX chain.

-g <vga1,vga2,vga3,vga4> to set gains individually for each TX chain.

By default: 0 is used for TX chains.

-w sets tx power.

-w <dBm> sets a common power in dBm for all TX chain.

-w <dBm1,dBm2,dBm3,dBm4> to set powers individually for each TX chain. Note -w and -g arguments are exclusive. If both are supplied, -g overwrite -w.

- b -b <bw> sets bandwidth in MHz. Default value is 20.
- m set MCSs to use. If multiple MCSs are specified, then the packets are sent with different MCSs in a round robin fashion
- m <mcs> to use a single MCS.
 - m <mcs1,mcs2,...,mcsn> to use a set of MCSs.
 - m <mcs1~mcs2> to use a continuous range of MCSs.
- By default: MCS 0 ~ MCS 15 are used.
- t -t <antenna mask> sets TX antenna mask. Accepted values are 0 ~ F, in hex format. Default value: F
- L -L selects low power PA. By default high power PA is selected.
- C -C to enable TX LO calibration. This overwrites the -o option.
- o -o <dc1_i,dc1_q,dc2_i,dc2_q,dc3_i,dc3_q,dc4_i,dc4_q> to set TX DC offset.
- k -k <baseband_tx_scaling_factor>. Range: 0 ~ 32. Default value is 16.
This factor is divided by 16 before it is applied to the output signal. Hence a factor of 16 provides a unit gain and a factor of 8 reduce the output power by 6 dB.
- s -s <tone index> send out a single tone. Valid tone index is -32 ~ 31.
When this option is specified, all other packets related parameters are discarded. The frequency of the tone is $40\text{MHz}/64 \times \text{tone_index}$ relative to the center frequency.
- i -i <inter frame spacing> set inter frame spacing, in uS. This is always set with respect to 20MHz bandwidth. It is scaled proportionally for other bandwidth. Set a minimum of 20. Default value: 20.
- It is between the time when the radio is put in TX mode and the packet is sent out. Default value is 0.25. This field corresponds to the 'RF_switch_delay' field in MAC registers 0x5, divided by 80.
- p -p <packet size> set MAC packet size, in 32-bit words. By default: 250 words.
- u -u <duty cycle> set the TX duty cycle. This argument is not applicable for tone transmission. The number supplied should be an integer and between 1 and 100. The actual duty cycle may not be the same as requested due to system constraints. When the duty cycle argument is provided, the packet length and inter-frame spacing arguments are only used when possible.
- a -a <aggregation size> set number of MAC packets to be aggregated into one PHY packet. By default: no aggregation
- n -n <number of PHY packets to send per MCS>. Use 0 for indefinite transmission. Default value: 0.

Example:

Using the argument as '-f 4450 -w 20 -t 1 -s 2' sends a tone with tx power = 20dBm, tx antenna #1, and on frequency (4450 + 1.25) MHz.

Frequency is a mandatory argument.

Note:

When entering values separated by comma or tilts, be sure there's no spaces in between.

3.38 Continuous Wave Stop

- **Command Deprecated**
- Command: cw_stop
- Parameter: N/A, this command does not need a parameter.
- Description: This command stops the operation invoked by running the "Continuous Wave Start" command. If the radio isn't currently running the Continuous Wave feature, an error message will return. After the Continuous Wave feature is stopped, all radio settings (frequency, bandwidth, etc) are returned to the values before running the Continuous Wave feature. The user is responsible to return the radio mode back to network mode (command 3.37).

3.39 Spectrum Scan Start

- Command: spectrum_scan_start
- Parameter: ["arguments"], where arguments is a string with several parameters.
- Description: This command starts spectrum scanning on the assigned frequencies. The arguments are explained below with an example in the end.

-f <freq> sets the center frequency, in Mhz.

-f <freq1,freq2> sets 2 separate frequencies

-f <freq1:step_size:freq2> sets a range of frequencies with step size equals to step_size.

The user can assign a combination of frequencies, such as
<freq1,freq2:step_size:freq3,freq4>.

The frequency span for scanning is fixed at 20MHz. Hence it is recommended the step-size is set to 20MHz or -20MHz for maximum scanning efficiency.

This argument is mandatory.

-a <antenna_mask> specifies which RX channel to scan. Valid values are 1 ~ F in hex format. For example a value of D means to scan RX channel 1 and 3 and 4. Default value is 1.

-c <capture_num> sets the number of captures for each frequency, should be an integer between 1 and 65535. Default value is 1.

-s <fft_size> sets the size for Fast Fourier Transform, this number should be a power of 2. A larger number will increase frequency resolution but takes longer to

compute. Resolution bandwidth (RBW) is 20MHz/fft_size. Valid range for fft_size is 1 ~ 512. Default value is 64 resulting a resolution bandwidth of 312.5KHz.

-g -g <input_power> sets the expected input power level, in dBm

-l -l <IP> sets the streaming report IP address, should be XXX.XXX.XXX.XXX, while XXX is a number between 0 and 255.

-p -p <port> sets the streaming report port, should be an integer between 0~65535, it is suggested to use a number larger than 2000.

-t -t <threshold> sets the threshold that the FFT output is compared against. The number of times the threshold is broken is recorded and reported. See Section 7 for more details. The unit of the threshold is dBm/Hz. Note theoretical white noise floor (KTB) at room temperature is about -174dBm/Hz.

A reasonable example for a quick scan of 4400MHz ~ 4700MHz would be:

"-c 20 -s 1 -i 192.168.10.220 -p 2007 -t -153 -f 4410:204690 -a 1". It will capture 20 times for each frequency mentioned, and perform no FFT. Threshold -153dBm/Hz represent about 15dB above noise floor assuming 5dB noise figure. The result will be sent to 192.168.10.220 at port 2007.

The result gathered by this command will be sent in a streaming response similar to RSSI reporting and temperature reporting, and will be detailed in section 7. Note that the user is responsible for returning the radio to the state before the scanning process, which is detailed in the "Spectrum Scan Stop" command.

- Write only API.

3.40 Spectrum Scan Stop

- Command: spectrum_scan_stop
- Parameter: Either none or 1
- Description: By default Spectrum Scan Start command will scan through all the specified frequencies and only quits when all the reports have been sent. Once the spectrum scan starts the radio will have all network functions disabled, even after the scan is complete.

The user can run this command without any parameters to restore the networking feature after a spectrum scan. If this command is invoked in the middle of spectrum scanning, it will return an error message.

If spectrum scanning is frozen or the user wish to terminate the scanning before its completion, the user can pass "1" as a parameter with this command. This will force quit the spectrum scanning and restore radio to the status before invoking the scan.

- Write only API.

3.41 Print All Radio Settings

- Command: print_all_settings

- Parameter: N/A, since this command does not need a parameter.
- Description: This command prints the radio settings in the form of a JSON array. The result format is:

```
["command1",["value1_1"],"command2",["value2_1","value2_2"],...]
```

"command" is a string indicating the variable name and should be the same name as the command names mentioned in section 3. The "value" is the corresponding value(s) of the variable, and is expressed as a JSON array. For example, the RSSI report address variable will be represented as follow:

```
[...,"rssi_report_address",["172.20.255.185","2000"],...]
```

- Read Output Format: ["freq",["2280"],"bw",["20"],"power_dBm",["10"],...], each pair of values corresponds to one API, the first value is a string depicting the API, the second value are the parameters attached to the API.

3.42 Burst Time

- Command: burst_time
- Parameter: ["time"], where time is an integer
- Description: The total on air time per packet transmission. Higher burst_time will increase throughput, due to readuced CSMA overhead, at the expense of latency.
- Unit: us
- Range: 2000~150000
- Default: 10000
- Read Output format: ["10000"], integer value in quotes.

3.43 GPS Coordinates

- Command: gps_coordinates
- Parameter: ["latitude","longitude","altitude"], where latitude, longitude and altitude are 3 floating point numbers. The numbers should represent a valid GPS coordinate.
- Description: This command returns the current GPS coordinates of the radio if no parameters are provided. The format is a JSON array with 3 floating point numbers. They represent Latitude, Longitude and Altitude, respectively. If the parameters are provided, it will set the node's position to those values.
- Range: Latitude: -90 - 90. Longitude -180 180. Altitude: -12400 - 42000.
- Default: 34.057, -118.447, 0
- Read Output format: ["34.057","-118.447","0"], floating point values in quotes.

3.44 External BDA mode

- Command: pa_mode
- Parameter: ["m"], where mode is an integer.

- Description: This command specifies the operational mode of the external BDA.
- Parameter is an integer which take the following values:
 - 0: No BDA attached.
 - 1: BDA attached in basic mode.
 - 2: BDA attached in advanced mode. This mode is currently not implemented.
- This API is readable and writeable.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.45 Basic BDA mode settings

- Command: pa_basic_mode_settings
- Parameter: ["settings"], where settings is a string of the following format: paGain1_paGain2_paGain3_paGain4_InaGain1_InaGain2_InaGain3_InaGain4_noiseFigure_MaximumPAOutputPower
- Description: This method takes a setting string, and replaces the current basic mode BDA settings with the new one. If no argument is given, it will just return the current setting string.
- All 10 parameters are necessary.
- The settings are only applicable if the BDA mode is set to basic mode using the pa_mode API command.
- paGain1-4 are the PA gains in dBm for channels 1-4.
- InaGain1-4 are the LNA gains in dBm for channels 1-4.
- noiseFigure is the noise figure of the BDAs in dBm (currently unused, user can put 0 as a placeholder).
- MaximumPAOutputPower is the maximum total output power that the BDA can handle. This is currently only used in the web interface to cap the radio output power in the Basic Settings page.
- **The user needs to take care and calibrate the radio output power using the "power_dBm" API command when the BDA is attached. If the radio output power that is being input to the BDA is too strong the BDA can get damaged.**
- Default: 3_3_3_3_4_4_4_4_0_31
- Range: value range 0-100
- Read Output format: ["3_3_3_3_4_4_4_4_0_31"], integer values separated by underscore in quotes

3.46 Automatic Noise Power Estimation

- **Deprecated**
- Command: noise_estimate_enable
- Parameter: ["enable"], where enable is an integer.
- Description: This command enables or disables automatic noise power estimation. Automatic noise power estimation is enabled if "enable" is set to 1, and disabled if "enable" is set to 0.

3.47 PA max output power per channel

- Command: `pa_basic_mode_max_output_power_per_channel`
- Parameter: ["power"], where power is an integer
- Unit: dBm
- Description: This command controls the maximum output power that the attached PA can support on a per channel basis. It is only used if the `pa_mode` is set to 1 (basic mode). If no parameter is passed, the current value is returned. If a parameter is passed, the parameter is set. The radio will cap its own total output power taking antenna mask into account to make sure that per-channel output power does not exceed this value e.g. if this value is set to 40 dBm, and the PA gain is 20 dBm, then the radio will never exceed its per-channel output power to be 20 dBm. If the antenna mask enables all channels, the radio will cap total output power to 26 dBm power (20 dBm per channel times 4). If the antenna mask enables 2 channels, it will cap total output power 23 dBm (20 dBm per channel times 2).
- Default: 1
- Range: 0-100
- Read Output Format: ["1"], integer value in quotes

3.48 Virtual IP Address

- Command: `virtual_ip_address`
- Parameter: ["vir_addr"], where vir_addr is the virtual IP address
- Description: This command will set the virtual IP address to be "vir_addr". If no parameter is given, it will show the current virtual IP address
- Default: 10.1.1.1
- Read Output Format: ["10.1.1.1"], string value in quotes

3.49 Virtual IP Network mask

- Command: `virtual_ip_netmask`
- Parameter: ["vir_netmask"], where vir_netmask is the virtual IP mask
- Description: This command will set the virtual IP mask to be vir_netmask. If no parameter is given, it will return the current mask.
- Default: 255.0.0.0
- Read Output Format: ["255.0.0.0"], string value in quotes

3.50 Virtual IP Gateway

- Command: `virtual_ip_gateway`
- Parameter: ["vir_gw"], where vir_gw is the virtual IP gateway
- Description: This command will set the virtual IP gateway to be vir_gw. If no parameter is given, it will show the current virtual IP gateway.
- Default: 10.1.1.2
- Read Output Format: ["10.1.1.2"], string value in quotes

3.51 Virtual IP disable

- Command: `virtual_ip_disable`
- Parameter: `["vir_ip_disable"]`, where `vir_ip_disable` is either 1 or 0
- Description: This command enable/disable virtual IP. If `vir_ip_disable` is set to 1, virtual IP will be enabled. Otherwise it will be enabled. If no parameter is given, it will return the current virtual IP status.
- Default: 1
- Read Output Format: `["1"]`, integer value in quotes

3.52 VPN IP Address

- Command: `vpn_address`
- Parameter: `["vpn_addr", "vpn_port"]`, where `vpn_addr` is the VPN IP address, and `vpn_port` is the VPN port number.
- Description: This command will set the VPN IP address and VPN port number to be `vpn_addr` and `vpn_port`. If no parameter is given, it will show the current VPN IP address with the VPN port number.
- Default: `["10.0.1.3", "9000"]`
- Read Output Format: `["10.0.1.3", "9000"]`, string value for ip and integer value for port in quotes

3.53 VPN Disable

- Command: `vpn_disable`
- Parameter: `["vpn_disable"]`, where `vpn_disable` is either 1 or 0
- Description: This command will disable VPN if `vpn_disable` is set to 1. Otherwise VPN will be enabled. If no parameter is given, it will show the current VPN status.
- Default: 1
- Read Output Format: `["1"]`, integer value in quotes

3.54 Local IP Address

- Command: `local_address`
- Parameter: NA
- Description: This command will show the local IP address of the radio.
- Read only API.
- Read Output Format: `["172.20.1.1"]`, string value in quotes

3.55 Queue Size

- Command: `queue_size`
- Parameter: `["node_id"]`
- Description: This command will show the transmit queue size of the node for a specific destination `"node_id"`. If there is no traffic, 0 will be the output.
- Read only API.
- Read Output Format: `["50"]`, integer value in quotes

3.56 Routing Beacon Period

- Command: `routing_beacon_period`
- Parameter: ["period"], where period is an integer
- Description: This command updates the radio's time period of sending routing beacons. This input setting is applied directly for 20MHz bandwidth and multiplied by 2.5 for 5 MHz bandwidth.
- Unit: ms
- Range: 100ms~2000ms
- Default: 100ms
- Read Output Format: ["12345"], integer value in quotes

3.57 Supported Frequencies

- Command: `supported_frequencies`
- Parameter: ["bandwidth", "ant_mask"], where bandwidth is a float, ant_mask is an integer bitmask. These are optional parameters.
- Range: bandwidth = 1.25, 2.5, 5, 10, 20. ant_mask = 0-15
- Description: This command returns the supported frequencies for the board in the same frequency format as the -f option in the Spectrum Scan Start command. The user needs to enter a bandwidth as the input parameter when using this command.
- Unit: mHz
- Read Output Format: ["2210","2220","2240"], list of frequency integer value in quotes

3.58 Read Power dBm

- Command: `read_power_dBm`
- Parameter: N/A, since this command is read only
- Description: This command returns the actual transmitted total output power in dBm
- Read Output Format: ["10"], integer value in quotes

3.59 Read Power mw

- Command: `read_power_mw`
- Parameter: N/A, since this command is read only
- Description: This command returns the actual transmitted total output power in mw
- Read Output Format: ["10"], integer value in quotes

3.60 Supported Frequency Profiles

- Command: `supported_frequency_profiles`
- Parameter: N/A, since this command is read only
- Description: This command returns the supported frequency/bandwidth/antenna mask combinations supported by this board. The

output is a JSON formatted array of objects. Each object contains a frequencies array in the format same as the `-f` option in the Spectrum Scan Start command, with the corresponding applicable bandwidth and antenna mask. If the bandwidth is `-1`, it is applicable for all supported bandwidths (5 and 20 MHz). Antenna mask is of the same format as defined in TX Antenna and RX Antenna commands.

- Read Output Format:

```
[
{"antenna_mask":"15","bandwidth":"-1",
"frequencies":["2200:20:2380","4700"]},
{"antenna_mask":"3","bandwidth":"-1",
"frequencies":["4420:40:4700"]},
{"antenna_mask":"12","bandwidth":"-1",
"frequencies":["4700:20:4980"]}
], combination of integer and string values.
```

3.61 Frequency Bandwidth

- Command: `freq_bw`
- Parameter: `["freq","bw"]`, where `freq` and `bw` are integers
- Description: This command allows the user to configure both the frequency and the bandwidth in a single API command, thus reducing the latency for setting both parameters individually. The corresponding unit and range is the same as Frequency and Bandwidth commands. This command is set only, also it can't be used with `setenvlinsingle`.
- Read Output Format: `["2280","20"]`, floating point values in quotes

3.62 Network Name

- Command: `nw_name`
- Parameter: `["network_name"]`, where `network_name` is a string
- Description: This command will return the Network ID without argument. The Network ID is similar in concept to the SSID used in WIFI. Only nodes programmed with the same Network ID will be able to mesh with each other. If a single string argument is provided, it will program the Network ID of the node to be this string.
- Default: `Silvus`
- This API is readable and writeable.
- Read Output Format: `["Silvus"]`, floating point values in quotes

3.63 Encryption Profile

- Command: `enc_profile`
- Parameter: `["algorithm","key_length"]`, where `algorithm` and `key_length` are integers.
- Description: This command will return the current encryption profile's algorithm and key length. To set the encryption profile, algorithm and key length must be provided. The mapping between different algorithm and key length is as follow:
0 128: AES ECB with 128 bit keys
0 256: AES ECB with 256 bit keys

28 64: DES ECB with 64 bit keys

26 256: AES GCM Static 256 bit keys

27 256: AES GCM ECDH with 256 bit keys

Any other combination of encryption algorithm and key length is incorrect and will result in undefined behavior.

- FIPS mode supports only these profiles: AES GCM Static 256-bit (26, 256) and AES GCM ECDH 256-bit (27, 256).
- This API is readable and writeable.
- Read Output Format: ["27"], integer value in quotes representing the encryption algorithm

3.64 Neighbor MCS

- Duplicate API. Same as 3.15

3.65 Neighbor Rx MCS

- Command: `nbr_mcs_rx`
- Parameter: ["nodeid"], where nodeid is an integer
- Description: This command takes one parameter - nodeid of a transmitter radio. It will return the MCS being used by the link adaptation algorithm on this transmitter radio to transmit to the local radio.
- Read only.
- Read Output Format: ["12"], integer value in quotes

3.66 Noise Level

- Command: `noise_level`
- Parameter: N/A, since this command is read only
- Description: When called on a radio, it will return the current noise level in dBm
- Read Output Format: ["29"], integer value in quotes

3.67 Neighbor RSSI

- Command: `nbr_rssi`
- Parameter: ["nodeid"], where nodeid is an integer
- Description: When called on a radio with parameter *nodeid*, it will return the current RSSI values from neighbor radio *nodeid* as an array of 4 values, channel 1 RSSI, channel 2 RSSI, channel 3 RSSI and channel 4 RSSI. If no data exists for neighbor *nodeid*, returned array will be empty.
- Unit: dBm
- Read Output Format: ["-38", "-48", "-41", "-45"], integer values in quotes

3.68 Routing Path

- Command: `routing_path`

- Parameter: ["nodeid"], where nodeid is an integer
- Description: When called on a radio with parameter *nodeid*, it will return the current routing path to the radio *nodeid* as an array of intermediate radio hops, e.g. a 2 hop route to *nodeid* would be ["H1", "H2", "*nodeid*"] where H1 is the local radio node id. If no route exists for *nodeid*, returned array will be empty.
- Read Output Format: ["22103","42902"], integer values in quotes

3.69 FPGA Build Info

- Command: fpga_build_info
- Parameter: N/A, since this command is read only
- Description: This command returns an array with PHY build tags: ["version", "git tag", "build date"]
- Read Output Format: ["1.12","0x6d8edf62d858be4a182dfac0bb54f497550af314","06-22-22, 11:02:10"], floating point version, string git tag hash and string build date in quotes

3.70 Enable Max Power

- Command: enable_max_power
- Parameter: ["enable"], where enable is either 1 or 0
- Description: If enabled, the radio will try to maximize the transmit power across all active channels. If BDA is enabled, the radio will still honor the "pa_basic_mode_max_output_power_per_channel". The transmit power set using "power_dBm" will be ignored. If disabled, the radio will use the "power_dBm" values to set the transmit power.
- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.71 Add SSH Key

- Command: add_ssh_key
- Parameter: ["key1"]
- Description: This is a write only command. This is used to authorize SSH access into the radio. The key pair can be generated like this, ``ssh-keygen -t ecdsa -b 521``. Replace key1 with the public key generated. This adds key1 to already existing keys.
- Write only API.

3.72 Remove SSH Key

- Command: remove_ssh_key
- Parameter: ["num"] where num is 0,1,2,...n
- Description: This removes the key added by add_ssh_key corresponding to num. If num is 0, it removes the first key, if num is 1, it removes second key, etc.
- Write only API.

3.73 List SSH Keys

- Command: `list_ssh_keys`
- Parameter: ["key1", "key2", "keyn"]
- Description: If no parameters are sent, this returns all the keys added by `add_ssh_key`. If parameters are specified, you overwrite all keys with the new keys in your array. Put a newline after each key in array. Example ["my_new_key\n", "my_other_key\n"].
- This API is readable and writeable.
- Read Output Format: ["my_new_key\n", "my_other_key\n"], list of string values in quotes

3.74 Generate Host Key

- Command: `generate_host_key`
- Parameter: N/A
- Description: Generates and sets a new SSH host key used for authenticating SSH connections. The key pair is elliptic curve secp521r1.
- Read Output Format: ["my_radio_private_key\n"], string value in quotes

3.75 Host Key

- Command: `host_key`
- Parameter: ["host_key"]
- Description: If no parameter is sent, this reads back the host key. If parameter is sent, it sets a new host key, replacing what was there before. A host key can be generated like this: ``openssl ecparam -name secp521r1 -genkey -noout -out myfile``.
- This API is readable and writeable.
- Read Output Format: ["my_radio_private_key\n"], string value in quotes

3.76 Generate TLS Certificate

- Command: `generate_tls_cert`
- Parameter: N/A
- Description: This is write only. This creates a new HTTPS certificate and replaces the old one.
- Read Output Format: ["my_radio_tls_cert\n"], string value in quotes

3.77 TLS Cert

- Command: `tls_cert`
- Parameter: ["your_tls_cert"]
- Description: If no parameter is sent, this will return the current HTTPS certificate. If parameter is sent, it will replace the current HTTPS certificate.
- This API is readable and writeable.
- Read Output Format: ["my_radio_tls_cert\n"], string value in quotes

3.78 RF Auth Key

- Command: `enc_rf_auth_key`
- Parameter: `["key1"]`
- Description: If no parameter is sent, this will return the current `enc_rf_auth_key`. Parameter `key1` is a 256-bit sequence, represented as 64 hex numbers. For this to do anything, the `api enc_profile` must be set to `["27", "256"]`.
- This API is readable and writeable.
- Read Output Format: `["1234657891..."]`, string value in quotes

3.79 FIPS Mode

- Command: `fips_mode`
- Parameter: `["mode"]` where `mode` is 1 for enable, 0 for disable
- Description: If no parameter is sent, this will return the current `fips_mode`. If parameter is sent, radio will reboot and will erase all setting profiles, reset the encryption key, both SSH keys, the HTTPS certificate, and the login passwords to their factory default. Enabling will also turn on HTTPS and Login Authentication.
- Default: 0
- This API is readable and writeable.
- Read Output Format: `["0"]`, integer value in quotes

3.80 Zeroize

- Command: `zeroize`
- Parameter: N/A
- Description: This will set login passwords and all security keys to their defaults. It will also erase all settings profiles. Also, if FIPS mode is off, it will turn off HTTPS and login.
- Note: starting with v4.0.3.0, this API will delete all existing users and create the 3 new users with usernames "admin", "basic" and "advanced" with the default password "HelloWorld", and access levels "admin", "basic" and "advanced" respectively.
- Write only.

3.81 Admin Pass

- Command: `admin_pass`
- Parameter: `["old password"]["new password"]`
- Description: This API will take one or two parameters depending on the value of the `"enable_password_complexity"` API (See 3.162). If `"enable_password_complexity"` is set to 0, it will take 1 parameter: the new password to be set, in plaintext.
- If `"enable_password_complexity"` is set to 1, it will take 2 parameters: the new password, and the old password. The complexity requirements (See 3.162)

will be checked before setting the new password. The radio will also check that the old password being passed matches what the radio expects.

- Only accessible by the admin user.
- Write-Only.
- Note: starting from v4.0.3.0, this API will
 - Change the password of the “admin” username if one exists OR
 - Automatically create a new username “admin” with access level “admin” and set the password.

3.82 Advanced Pass

- Command: advanced_pass
- Parameter: [“old password”][“new password”]
- Description: Same as admin_pass, but for the Advanced user.
- Only accessible by the admin user.
- Note: starting from v4.0.3.0, this API will
 - Change the password of the “advanced” username if one exists OR
 - Automatically create a new username “advanced” with access level “advanced” and set the password.
- Write-Only.

3.83 Basic Pass

- Command: basic_pass
- Parameter: [“old password”][“new password”]
- Description: Same as admin_pass, but for the Basic user.
- Only accessible by the admin user.
- Note: starting from v4.0.3.0, this API will
 - Change the password of the “basic” username if one exists OR
 - Automatically create a new username “basic” with access level “basic” and set the password.
- Write-Only.

3.84 PTT Status

- Command: ptt_status
- Parameter: [“num”] where num is either 0 or 1. 1 for enable
- Description: **In order for any of the following PTT API’s to take effect, this command must be toggled, (turn off, then on).** This enables/disables PTT.
- Default: 1
- Read Output Format: [“1”], integer value in quotes

3.85 PTT Speaker Volume

- Command: ptt_speaker_vol
- Parameter: [“num”] where num is 0-100

- Description: This adjusts gain on the speaker. 100 for max gain.
- Default: 80
- Read Output Format: ["80"], integer value in quotes

3.86 PTT Mic Bias

- Command: ptt_mic_bias
- Parameter: ["num"] where num is either 0 or 1. 0 is for 3 Volts, 1 is for 2.15 Volts.
- Description: This represents the bias voltage that will go to the PTT module.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.87 PTT Mic Volume

- Command: ptt_mic_vol
- Parameter: ["num"] where num is 0-100.
- Description: This represents the gain on the microphone. 100 for max gain.
- Default: 80
- Read Output Format: ["80"], integer value in quotes

3.88 PTT Mic Type

- Command: ptt_mic_type
- Parameter: ["num"] where num is either 0 or 1. 0 is for coil, 1 is for condenser
- Description: This represents the type of microphone being used.
- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.89 PTT Audio Encoder Type

- Command: ptt_audio_encoder_type
- Parameter: ["num"] where num is either 0, 1, or 2. 0 for G.722 (high quality), 1 for G.711, 2 for Variable Rate Codec.
- Description: Selects the codec. (Also see opus_rate command)
- Default: 2
- Read Output Format: ["2"], integer value in quotes

3.90 Opus Rate

- Command: opus_rate
- Parameter: ["num"] where num is 4000 – 100000. The units are bits per second.

- Description: This command is only relevant if ptt_audio_encoder_type is "2". This controls the audio codec rate.
- Default: 30000
- Read Output Format: ["30000"], integer value in quotes

3.91 PTT Mcast Group

- Command: ptt_mcast_group
- Parameter: ["index", "IP"] where index is 0-15 and represents the Multicast Groups 1-15 on the GUI. IP is a multicast IP address.
- Description: Radios will only communicate with radios with the same multicast address. To read the value of this command, provide the index. For example, "ptt_mcast_group 0". (See also ptt_active_mcast_group).
- Default: ["0", "239.0.0.190"]
- Read Output Format: ["239.0.0.190"], string ip value in quotes

3.92 PTT Active Mcast Group

- Command: ptt_active_mcast_group
- Parameter: ["r₀,r₁...r₄_t₀,t₁...t₄_m₀,m₁...m₄"] Variables r_i, t_i, m_i are numbers 0-15. They represent indexes to the ptt multicast groups. r_i stands for listen, t_i stands for talk, and m_i stands for monitor.

- Default: 0_0_

- Description: This one is a little confusing. This API controls the "Inactive," "Active," "Monitor" dropdown on the PTT GUI.

Active - To make group 0 and 1 active: "0,1_0,1". (Note there must always be at least 1 active group).

Monitor – To make group 0 and 1 monitor and 2 active: "0,1,2_2_0,1". You must set r_i and m_i for monitor.

Inactive – Simply remove required index from the string. In the monitor example, 3 and 4 are inactive.

- Read Output Format: ["0_0"], list of integer values separated by underscore in quotes

3.93 PTT Beep Volume

- Command: ptt_beep_volume
- Parameter: ["num"] where num is 0 -100.
- Description: This controls beep volume that happens when you try to talk while another user is speaking.
- Default: 100
- Read Output Format: ["100"], integer value in quotes

3.94 PTT Aggr Delay

- Command: ptt_aggr_delay

- Parameter: ["num"] where num is 120 -360. Measured in milliseconds.
- Description: Lower values will have lower latency, higher values will have higher efficiency.
- Default: 180
- Read Output Format: ["180"], integer value in quotes

3.95 PTT HQ Link Notifications

- Command: ptt_hq_link_notifications
- Parameter: ["nodeid"] where nodeid is nodeid of hq radio. -1 for disable.
- Description: Corresponds to PTT HQ Link Notifications on GUI.
- Default: -1
- Read Output Format: ["-1"], integer value in quotes

3.96 PTT Levels Link Notifications

- Command: ptt_levels_link_notifications
- Parameter: ["x1_x2_x3_x4"] where x_i is 0-70.
- Description: Configures the number of link levels. If you want two link levels, say between 0-20dB and 20-70dB, your string would look like "20". For three link levels, it might look like "20_40". For five, "10_20_30_40". Supports up to 10 levels
- Default: 10_20
- Read Output Format: ["10_20"], list of integer values separated by underscore in quotes

3.97 PTT Level Change Link Notifications

- Command: ptt_level_change_link_notifications
- Parameter: ["num"] where num is either 0 or 1. 1 for enable.
- Description: Automatically notifies you on link level transition.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.98 PTT Notification Volume

- Command: ptt_notification_volume
- Parameter: ["num"] where num is 0-100.
- Description: Controls notification volume.
- Default: 100
- Read Output Format: ["100"], integer value in quotes

3.99 PTT No Link Notifications

- Command: ptt_no_link_notifications
- Parameter: ["num"] where num is either 0 or 1. 1 for enable

- Description: Repeat notifications when no link to HQ.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.100 IGMP Snoop Enable

- Command: igmp_snoop_enable
- Parameter: ["num"] where num is either 0 or 1. 1 for enable
- Description: Enable/disable igmp snooping, (multicast).
- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.101 IGMP Mesh Forwarding

- Command: igmp_snoop_mesh_forwarding_mode
- Parameter: ["num"] where num is either 0 or 1. 1 for "Block" and 0 for "Send to all radios"
- Description: Controls local and mesh forwarding behavior for multicast traffic that has no igmp snooping entries.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.102 Custom Pruning / Augmenting

- Command: custom_mcast_filter_enable
- Parameter: ["num"] where num is either 0 or 1. 1 for enable
- Description: Enable/disable custom pruning (multicast).
- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.103 Multicast Config

- Command: mcast_config
- Parameter: ["index", "IP"] where index is 0-4 and represents the Multicast Groups 1-5 on the GUI. IP is a multicast IP address.
- Description: Multicast_ip_address, receiver_id1, receiver_id2, ... , receiver_idN. If IGMP snooping is disabled, multicast traffic will only be forwarded to the radios in this list. If enabled, multicast traffic will only be forwarded to radios in this list that have client devices requesting this traffic. Traffic may be forced to go to a radio by adding the node with postfix "+". Traffic may be prevented from reaching a radio by adding postfix "-". E.g. "224.50.50.50 1234, 1235+, 1236-" - If snooping is disabled, 1234 and 1235 will receive traffic. If enabled, 1234 will ONLY receive traffic if connected clients ask for it. 1235 will always receive traffic and 1236 will never receive traffic. All other radios will NOT receive traffic. If receiver_id is -1, it will stop multicast traffic for this group, e.g. to stop all traffic for group 224.50.50.50, set it to "224.50.50.50, -1".

- Read Output Format: ["224.50.50.50, 1234, 1235+, 1236-"], comma-separated string values in quotes

3.104 Local Broadcast

- Command: mcast_xor_disable
- Parameter: ["num"] where num is either 0 or 1. 1 for disable
- Description: Enable/disable local broadcast (multicast).
- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.105 Local Broadcast MCS

- Command: mcast_xor_mcs
- Parameter: ["mcs"], where mcs is an integer. 0-15, 255 for auto
- Description: Sets/reads the MIMO modulation and coding rate.
- Default: 0
- Read Output Format: ["1"], integer value in quotes

3.106 Local Broadcast Fragmentation Threshold

- Command: mcast_xor_aggr_thresh
- Parameter: ["thresh"], where thresh is an integer.
- Description: Sets/reads the fragmentation threshold. An incoming packet of the size larger than the specified threshold is fragmented.
- Unit: byte
- Range: 200, 400, 800, 1600
- Default: 1600
- Read Output Format: ["1600"], integer value in quotes

3.107 Local Broadcast Multicast Groups

- Command: mcast_xor_group_list
- Parameter: List of Multicast IPv4 addresses separated by comma (,), e.g. 224.50.50.50, 224.50.50.51. Multicast addresses should only be separated by a single comma.
- Description: Traffic for these multicast groups will be sent using Local Broadcast.
- Default: ""
- Read Output Format: ["224.50.50.50, 224.50.50.51-"], comma-separated string values in quotes

3.108 Local Broadcast Rate Mode

- **Command Depreciated**
- Command: mcast_xor_rate_mode
- Parameter: ["num"] where num is either 0 or 1. 0 for "Auto" and 1 for "Fixed".

3.109 Local Broadcast Block Size

- **Command Deprecated**
- Command: `mcast_xor_block_size`
- Parameter: ["num1_num2"] where num1 is Rows and num2 is Columns.

3.110 Local Broadcast Target Latency

- Command: `mcast_xor_target_latency`
- Parameter: ["num"] where num is between 10 and 1000. Units in milliseconds.
- Default: 100
- Read Output Format: ["100"], integer value in quotes

3.111 Weakest Link

- Command: `weakest_link`
- Parameter: ["node_id"] where node_id is a number between 0 and 65535 representing the destination node id.
- Description: Returns the weakest link in the path to the destination node in the form of: <link source node id>, <link destination node id>, <link snr in dB>, <reuse factor>.
- The "reuse factor" is a number between 0-3 which represents the largest consecutive number of wireless links along the path that also includes the weakest link. This information is then used to probe the link throughput using the "Link Throughput" API.
- Read only.
- Read Output Format: ["22103", "42902", "37", "1"], comma-separated integer values in quotes

3.112 Link Throughput

- Command: `link_throughput`
- Parameter: ["node_id", "reuse_factor"] where node_id is a number between 0 and 65535 representing the node id of the destination of the link; the node that the API is invoked on is the source. The reuse factor is a number between 0-3 which represents the largest consecutive number of wireless links along the path that also includes the weakest link. See API command "Weakest Link".
- Description: Returns the UDP throughput in Mbps for a given reuse factor on a link that is returned by the API command "Weakest Link". The API is invoked on the source of the link, and the destination of the link is passed as the parameter.
- e.g. Consider a network with topology A-B-C-D, where A,B,C and D are the node ids. The user wants to probe the throughput on this path A->D. He will first run command "Weakest Link" on A with parameter D. Say it returns

["C","D",3], which tells us that C->D is the weakest link on the path, and the reuse factor is 3. Then invoke API "Link Throughput" on C with parameters D and 3. This will return the throughput in Mbps which will be the throughput for the entire route A->D.

- Read Output Format: ["5.129452"], floating point value in quotes

3.113 Guard Interval (GI) Mode

- Note: There are 3 separate commands that need to be used to set the GI Mode

- Command: `la_mode`
- Parameter: ["num"] where num is 1-3. An `la_mode` of 1 specifies a "Fixed" GI mode while an `la_mode` of 2 specifies a "Variable" GI mode. Setting this to 3 is a special "Variable" GI Mode which adapts MCS 0-31 whereas modes 1 and 2 only adapt MCS 0-15.
- Read Output Format: ["2"], integer value in quotes
- Command: `var_gi_mode`
- Parameter: ["num"] where num is 0 or 3. A `var_gi_mode` of 0 should only be used for a GI mode of "Fixed-16". The rest of the modes use 3
- Read Output Format: ["3"], integer value in quotes
- Command: `var_gi_cyc_pre_len`
- Parameter: ["num"] where num is in the set $S=\{16, 24, 32, 40, 48, 56, 63\}$
- Default: `var_gi_mode = 2`, `var_gi_mode = 3`, `var_gi_cyc_pre_len = 32`
- Examples:
 - Fixed - 16 => `la_mode = 1`, `var_gi_mode = 0`, `var_gi_cyc_pre_len = 16`
 - Fixed - 24 => `la_mode = 1`, `var_gi_mode = 3`, `var_gi_cyc_pre_len = 24`
 - Variable - 16/24 => `la_mode = 2`, `var_gi_mode = 3`, `var_gi_cyc_pre_len = 24`
 - Variable - 16/32 => `la_mode = 2`, `var_gi_mode = 3`, `var_gi_cyc_pre_len = 32`
- Read Output Format: ["32"], integer value in quotes

3.114 MAN-IA Config

- Command: `stdsa_freq_config`
- Parameter: ["enable", "num_valid_freq", "NVF1", "NVF2", ...] where *enable* is 1 for enabling MAN-IA. *num_valid_freq* is a number between 2 and 6 representing the freqs the engine will monitor. *NVF1* – *NVF5* are the alternate freqs the radio monitor. These freqs must be supported by the radio. *NVF0* is the configured frequency by the "freq" API.
Example: ["1", "4", "2220", "2240", "2260"]
- Default: ["0", "2", "", "", "", "", ""]

- Read Output Format: ["0", "2", ",", ",", ",", ",", ","], comma-separated integer values in quotes

3.115 MAN-IA Freq Selection Mode

- Command: stdsa_camp_freq_selection_mode
- Parameter: ["mode"] where *mode* is 0 for auto and 1 for fixed. Auto means the radio will intelligently monitor all alternate freqs and jump to the best freq. Fixed means it will stay on the freq given by the index configured in stdsa_camp_freq_fixed_index api below.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.116 MAN-IA Freq Fixed Index

- Command: stdsa_camp_freq_fixed_index
- Parameter: ["index"] where *index* is 0 - (num_valid_freq -1). The radio will stay on this freq index if stdsa_camp_freq_selection_mode is set to 1 (fixed).
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.117 MAN-IA Recommended Freq

- Command: stdsa_recommended_frequency
- Read Only. This api returns the freq index the MAN-IA engine recommends. This api corresponds to the lime green highlighting on the web GUI.
- Read Output Format: ["0"], integer value in quotes

3.118 MAN-IA Table

- Command: stdsa_scan
- Read Only. This api returns info that is shown on the bar graph and table of the web GUI.
- Read Output Format: [{"freq": "2310", "info": [{"nodeid": "12345", "rssi": "30", "age": "34"}]}, {"freq": "2320", "info": [...]}], json array of frequencies and their associated RSSI values per node in the "info" dictionary.

3.119 Voltage Input Monitoring

- Command: input_voltage_monitoring
- Read Only. Units in millivolts. This is the voltage displayed on the GUI header.
- Read Output Format: ["12197.363281"], floating point value in quotes

3.120 GPS Lock

- Command: `gps_mode`
- Read Output Format: ["unlocked"], string value in quotes
- Read Only. Shows different levels of gps lock.
 - unlocked: The gps device is not connected or has some other issue.
 - locked-1d: The connected gps device does not have a fix. NMEA data is available.
 - locked-2d: Gps device has lock with latitude and longitude data.
 - locked-3d: Gps device has lock with latitude,longitude, and altitude data.

3.121 Beamforming

- Command: `beamform_enable`
- Parameter: ["enable"] where *enable* is 1 for on and 0 for off.
- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.122 Battery Percentage

- Command: `battery_percent`
- Read-only. Divide response by 100 to return the battery percentage shown in the GUI.
- Read Output Format: ["65.0"], floating point value in quotes

3.123 Routing Mode

- Command: `routing_proto`
- Parameter: ["mode"] where options are "Legacy" and "path_vector" for the new algorithm. "Path_vector" is the default for SS4. No license is required.
- Default: `path_vector`
- Read Output Format: ["path_vector"], string value in quotes

3.124 Build Tag

- Command: `build_tag`
- Reads back the build tag of the radio.
- Read-only.
- Read Output Format: ["v4.0.3.16rc16"], string value in quotes

3.125 Build Information

- Command: `build_information`
- Reads back the build information of the radio, including radio Model type.
- Read Output Format:

{

```

"board_type":"SC4044",
"rfboard":"SN 2721 opt 21 range 2200~2500 4400~4940 ",
"rfboard1":"SN 2726 opt 21 range 2200~2500 4400~4940 ",
"digital_board":"SN 19112",
"model":"SC4400E Rev B1",
"build_date":"10-10-2019",
"build_tag":"streamscape_v4.1.0.0rc54_sl4k",
"phy_ts":"EX_1.12_0x6d8edf62d858be4a182dfac0bb54f497550af314_06-22-22,
11:02:10",
"part_no":"SC4480E-235467-SBST",
"entropy_version":"Silvus Clock Jitter Entropy Module v1.0"
}

```

Json dictionary of string values. The firmware version is represented by the json object "build_tag" and the hardware part numbers by "part_no" and "model".

- Read-only.

3.126 Retransmissions

- Command: num_retx
- Parameter: ["value"] Value of >=20 enables, 1=disabled.
- Enables or disable retransmissions for QoS normal queue, by default all traffic goes into this queue.
- Default: 20
- Read Output Format: ["20"], integer value in quotes

3.127 High Priority Retransmissions

- Command: num_retx_hp
- Parameter: ["value"] Value of >=20 enables, 1=disabled.
- Enables or disable retransmissions for QoS high priority queue.
- Default: 20
- Read Output Format: ["20"], integer value in quotes

3.128 Critical Priority Retransmissions

- Command: num_retx_hp2
- Parameter: ["value"] Value of >=20 enables, 1=disabled.
- Enables or disable retransmissions for QoS critical queue.
- Default: 20
- Read Output Format: ["20"], integer value in quotes

3.129 RTS Retries

- Command: num_rts_retx
- Parameter: ["value"] Value 1 through 4.
- Number of RTS retransmissions.

- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.130 Routing Beacon MCSz

- Command: routing_beacon_mcs
- Parameter: ["mcs"] where mcs is an integer.
- Sets/reads the MIMO modulation and coding rate
- Range: 0-6, 8-14
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.131 LED Brightness

- Command: led_brightness
- Parameter: ["brightness"] brightness between 0 to 15 (0 for maximum brightness and 15 for minimum brightness)
- Default: 4
- Read Output Format: ["4"], integer value in quotes

3.132 GPS Time

- Command: gps_time
- Reads back the gps time of a connected GPS device in seconds.
- Read-only.
- Read Output Format: ["4561387.1654"], floating point value in quotes

3.133 LED Toggle

- Command: led_disable
- Parameter: ["toggle"] where toggle is 0 to Enable, 1 to Disable.
- This API is readable and writeable.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.134 Whitelist/Blacklist Selection

- Command: wb_list_type
- Parameter: ["selection"] where selection is 0=disable, 1=white_list, 2=black_list
- Disables or enables the whitelist or blacklist specified by the white_list or black_list APIs.
- This API is readable and writeable.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.135 Whitelist

- Command: white_list
- Parameter: ["nodeid"+] : a list of one or more nodeids
- Example: ["2838", "8503"]
- Radio will only communicate with the radios matching these nodeids.
- This API is readable and writeable.
- Default: [""]
- Read Output Format: ["2838", "8503"], comma separated list of integer values in quotes

3.136 Blacklist

- Command: black_list
- Parameter: ["nodeid"+] : a list of one or more nodeids
- Example: ["2838", "8503"]
- Radio will not communicate with the radios matching these nodeids.
- This API is readable and writeable.
- Default: [""]
- Read Output Format: ["2838", "8503"], comma separated list of integer values in quotes

3.137 Node Label

- Command: node_label or node_labels
- Returns a list of all saved node labels on the radio and their corresponding nodeid.
- This API is read-only.
- Read Output Format: {"2838":"node1", "8503":"node2"}, json dictionary of nodes and their corresponding label

3.138 Infrastructure Mesh Node Type

- Command: mesh_node_type
- Description: Use this to set the infrastructure and mobile node types
- Parameter: ["node_type"] Values: 0 - Mesh, 1 - Infrastructure, 2 - Edge (Mobile)
- This API is readable and writeable.
- Default: 0 – Mesh
- Read Output Format: ["0"], integer value in quotes

3.139 Infrastructure Mesh Scan List

- Command: mobile_mesh_scan_list
- Parameter: An array of one or more ["freq", "bw"] entries. This will be the frequency scan list.

- Default: If left blank, the radio will scan the frequencies from the supported_frequencies API. Bandwidth will not change; the current configured bandwidth will be used.
- Example: `[["2240", "20"], ["2260", "20"]]`
- This API is readable and writeable.
- Default: `[]`
- Read Output Format: `[["2240", "20"], ["2260", "20"]]`, json array of frequency and bandwidth floating point pairs

3.140 Infrastructure Mesh Scan Result

- Note: Will be available in a future release.
- Command: `mobile_mesh_scan_result`
- Description: After the scan has completed, this list will be generated. This API returns a list of infrastructure radios scanned in the following format: `[[freq, bw, nodeid, network_id, SNR_detected], ...]`
- Read-only
- Read Output Format: `[["2240", "20", "15648", "38202", "45"], ["2260", "20", "15648", "38202", "30"]]`, json array of frequency and bandwidth floating point, and node id, network id, and SNR integer values

3.141 Local Broadcast Mode

- Command: `mcast_mhop_mode`
- Parameter: `["mode"]` 0 for single-hop, 2 for multi-hop
- When set to single-hop, `mcast_xor_group_list` traffic will flow using the local multicast algorithm for one hop, the rest of the hops will use the default multicast algorithm. When set to multi-hop, all hops will follow the local multicast algorithm.
- Default: 0
- Read Output Format: `["0"]`, integer value in quotes

3.142 Local Broadcast Single Receiver Optimization

- Command: `mcast_one_receiver_enable`
- Parameter: `["mode"]` 0 for minimum of 2 receivers, 1 for minimum of 1 receiver
- Specifies the minimum number of downstream receivers required for triggering the local multicast algorithm on the `mcast_xor_group_list` list of multicast groups.
- Default: 0
- Read Output Format: `["0"]`, integer value in quotes

3.143 Local Statistics – Input Unicast

- Command: `local_stats_input_uc`
- Input unicast rate for the local radio in Mbps. This is the same statistics visible in the GUI section: "Network Management" -> "Table View"

- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.144 Local Statistics – Input Broadcast/Multicast

- Command: local_stats_input_bc_mc
- Input broadcast and multicast rate for the local radio in Mbps. This is the same statistics visible in the GUI section: "Network Management" -> "Table View"
- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.145 Local Statistics – Input Dropped

- Command: local_stats_input_dropped
- Input drop rate for the local radio in Mbps. This is the same statistics visible in the GUI section: "Network Management" -> "Table View"
- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.146 Local Statistics – Forwarded Unicast

- Command: local_stats_fw_uc
- Forwarded unicast rate for the local radio in Mbps. This is the same statistics visible in the GUI section: "Network Management" -> "Table View"
- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.147 Local Statistics – Forwarded Broadcast/Multicast

- Command: local_stats_fw_bc_mc
- Forwarded Broadcast and Multicast rate for the local radio in Mbps. This is the same statistics visible in the GUI section: "Network Management" -> "Table View"
- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.148 Local Statistics – Interference

- Command: local_stats_interference
- Local radio interference in dB. This is the same statistics visible in the GUI section: "Network Management" -> "Table View"
- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.149 Link Statistics – Total Air Time

- Command: link_stats_total_air_time
- Parameter: ["nodeid"] – the nodeid of the neighbor node
- This is the total air time per neighbor link, the value returned includes all traffic types (unicast, broadcast, multicast). This is the same statistics visible in the GUI section: "Network Management" -> "Network Topology" -> "Link Statistics" (click on the individual radio links to view the "Link Statistics" popup).
- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.150 Link Statistics – Total Data Rate

- Command: link_stats_total_rate
- Parameter: ["nodeid"] – the nodeid of the neighbor node
- This is the total data rate in Mbps per neighbor link, the value returned includes all traffic types (unicast, broadcast, multicast). This is the same statistics visible in the GUI section: "Network Management" -> "Network Topology" -> "Link Statistics" (click on the individual radio links to view the "Link Statistics" popup).
- Read-only
- Read Output Format: ["\0.3423\"], floating point value in quotes

3.151 Link Statistics – Total Unicast Data Loss

- Command: link_stats_total_loss
- Parameter: ["nodeid"] – the nodeid of the neighbor node
- This is the total unicast data loss percent per neighbor link. This is the same statistics visible in the GUI section: "Network Management" -> "Network Topology" -> "Link Statistics" (click on the individual radio links to view the "Link Statistics" popup).
- Read-only

- Read Output Format: ["32"], integer value in quotes

3.152 Link Statistics – Unicast Variable GI Mode

- Command: link_stats_var_gi_mode
- Parameter: ["nodeid"] – the nodeid of the neighbor node
- Returns fixed or variable for the specific link. This is the same statistics visible in the GUI section: "Network Management" -> "Network Topology" -> "Link Statistics" (click on the individual radio links to view the "Link Statistics" popup).
- Read-only
- Read Output Format: ["3"], integer value in quotes

3.153 Link Statistics – Unicast Variable GI Length

- Command: link_stats_var_gi_cyc_pre_len
- Parameter: ["nodeid"] – the nodeid of the neighbor node
- Returns Guard Interval currently in use by the specified link, 16-63. This is the same statistics visible in the GUI section: "Network Management" -> "Network Topology" -> "Link Statistics" (click on the individual radio links to view the "Link Statistics" popup).
- Read-only
- Read Output Format: ["32"], integer value in quotes

3.154 Encryption Key Volatile

- Command: enc_key_volatile_disable
- Parameter: ["toggle"] – 0 for enabled, 1 for disable
- When set to enable, encryption settings and keys will not be saved to flash; they will be removed upon reboot. If disabled, encryption settings and keys will be able to be saved to the flash so they will persist across reboots.
- Default: 1
- Read-write
- Read Output Format: ["1"], integer value in quotes

3.155 Query Settings Profiles

- Command: list_of_profiles
- Will return the names of the settings profiles on the radio in a string separated by a new line character "\n"
- Return Example: Profile01\nProfile02\nProfile03
- Read-only
- Read Output Format: ["Profile01\nProfile02\nProfile03"], newline ('\n') separated list of string profile names in quotes

3.156 Read Settings Profiles

- Command: get_profile
- Parameter: ["profile_name"] – the exact name of the setting profile

- Will return the full contents of the settings profile specified.
- Read-only
- Read Output Format:
["{\n\"version\":\n\"0.3\", \n\"settings\":{\n{\n\"method\":\n\"max_link_distance\", \n\"id\":3, \n\"params\":{\n\"5000\" \n} \n}, \n{\n\"method\":\n\"enable_max_power\", \n\"id\":4, \n\"params\":{\n\"0\" \n} \n}, \n....}"]], json object containing the profile, enclosed in quotes

3.157 MPS Enable

- Command: mps_disable
- Parameter: ["toggle"] – 0 to enable, 1 to disable
- Enables or disables the MPS feature. When enabled, the MPS position will switch between the different sets of settings (these are configured via the mps_profile API).
- Default: 1
- Read-write
- Read Output Format: ["1"], integer value in quotes

3.158 MPS Supported

- Command: mps_supported
- Returns 1 if MPS is supported on the radio, else returns 0.
- Read-only
- Read Output Format: ["1"], integer value in quotes

3.159 MPS Position

- Command: mps_pos
- Reads back the current position of the MPS.
- Read-only
- Read Output Format: ["1"], integer value in quotes

3.160 MPS Frequency

- Command: mps_freq
- Optional Parameter: ["mps_position"] – mps position 0-13
- If no parameter is provided, will return the frequency configured on the current MPS position. If the MPS position is specified, will return the frequency configured on that MPS position.
- Read-only
- Read Output Format: ["2230"], floating point value in quotes

3.161 MPS Profile

- Command: mps_profile
- Read: will return all the settings for each position as a json string.

- **Write:** Requires that you send the entirety of the current settings. For example, if you want to add a new position, you will need to include all the current positions and the new position as parameters to the API. To edit an existing profile, you would need to retrieve the current settings and make the edits to the position desired. To set the group, you would include passing in the "group" parameter to the mps_profile. "Group" should be "network", "volume", or "talk group". "network" selects the "nw_name", "freq", and "bw" for the current MPS position. "volume" selects the "ptt_speaker_vol" for the current MPS position. "talk group" selects the "ptt_mcast_group" for the current MPS position. See below for an example which follows the syntax of a saved json settings profile.

- **Read Output Format:**

```
{
  "1": {
    "bw": "10",
    "freq": "2240",
    "nw_name": "Silvus1",
    "ptt_mcast_group": "239.0.0.190",
    "ptt_speaker_vol": "80"
  },
  "2": {
    "bw": "10",
    "freq": "4440",
    "nw_name": "Silvus1"
  },
  "3": {
    "bw": "10",
    "freq": "4405",
    "nw_name": "Silvus1"
  },
  "4": {
    "bw": "10",
    "freq": "2290",
    "nw_name": "Silvus1"
  },
  "5": {
    "bw": "10",
    "freq": "4400",
    "nw_name": "Silvus1"
  },
  "6": {
    "bw": "10",
    "freq": "4420",
    "nw_name": "Silvus1"
  },
  "7": {
```



```

    "bw": "10",
    "freq": "2240",
    "nw_name": "Silvus2"
  },
  "group": "network",
  "group_types": {
    "network": {
      "freq_bw": 1,
      "nw_name": 1
    },
    "talk_group": {
      "ptt_mcast_group": 1
    },
    "volume": {
      "ptt_speaker_vol": 1
    }
  },
  "version": "1.0"
}, json dictionary object with string value pairs
• Read-write

```

3.162 Password Complexity

- Command: enable_password_complexity
- Parameter: ["toggle"] – 1 for enabled, 0 for disable
- When set to enable, any new password changes will require the user to pass in the old password. Then the radio will validate that the complexity requirements have been met.
 - Complexity Requirements: The new password must have at least 1 lowercase, 1 uppercase, 1 digit and 1 special character. It also applies configured minimum password length (See below) and minimum password change settings (See below) to new passwords.
 - 32 special characters are allowed: @#!~\$%^&*()-+/:.,<>?|_[]="'\{};`
 - When set to be disabled, the radio will not require the old password for password changes, and no complexity requirements will be checked.
 - Default: 0
 - Read-write
 - Read Output Format: ["0"], integer value in quotes

3.163 Minimum Password Length

- Command: min_password_length
- Parameter: ["length"] – minimum length of the password.

- This API only applies when enable_password_complexity (See above) is enabled. Any new password changes will need to be of this minimum length.
- Default: 5
- Range: 1-64
- Read-write
- Read Output Format: ["5"], integer value in quotes

3.164 Minimum Password Change

- Command: min_password_change
- Parameter: ["value"] – minimum number of characters that must be changed when compared to the old password.
- This API only applies when enable_password_complexity (See above) is enabled. The radio will validate that the new password being set is different from the old password in at least the minimum number of characters set here.
- Default: 3
- Range: 1-min_password_length
- Read-write
- Read Output Format: ["3"], integer value in quotes

3.165 Enable API Logging

- Command: enable_api_logs
- Parameter: ["value"] – Enable (1) or Disable (0).
- After enabling this API, the radio will log all API actions and the result (Success or Failure) to a log file. The log file can be downloaded from the GUI, or log entries can be sent in real-time to a Syslog server (see below)
- Default: Disable (0)
- Read-write
- Read Output Format: ["0"], integer value in quotes

3.166 Log Server

- Command: send_log_to
- Parameter: ["IP"] – IP is the IPv4 address of a Syslog server.
- All log entries can be sent to a remote Syslog server in real-time.
- Default: Empty (no log to server)
- Read-write
- Read Output Format: ["172.20.11.11"], ip string value in quotes

3.167 Notice of Use

- Command: notice_of_use
- Parameter: ["Notice Content"] – Put the entire content as text here.
- This API allows the Admin to set the Notice of Use/Banner upon first access to the Web GUI.
- Default: "Unauthorized or improper use of this system may result in administrative disciplinary action, civil charges/criminal penalties, and/or other sanctions. By continuing to use this system you indicate your awareness of and consent to these terms and conditions of use."
- Read-write
- Read Output Format: ["Unauthorized or improper use of ..."], string value in quotes

3.168 Notice of Use (Console)

- Command: console_notice_enable
- Parameter: ["value"] – Enable (1) or Disable (0)
- This API allows the Admin to enable display of the Notice of Use on the serial console.
- Default: Disable (0)
- Read-write
- Read Output Format: ["0"], integer value in quotes

3.169 Disable Concurrent Sessions

- Command: disable_concurrent_sessions
- Parameter: ["value"] – Disable (1) or Enable (0)
- This API allows the Admin to enable or disable concurrent sessions, e.g. if set to 1, a user cannot authenticate to the "login.sh" endpoint more than once concurrently. This will limit each user to only 1 browser session. After log in, the user needs to explicitly log out of the session using the "logout" API, or allow the session to expire automatically after the idle timeout.
- Default: Enabled (0)
- Read-write
- Read Output Format: ["0"], integer value in quotes

3.170 Max Number of Authentication Attempts

- Command: max_auths
- Parameter: ["value"] – Maximum number of consecutive failed authentication attempts before lock out.
- This API allows the Admin to set the maximum number of times a user can log in unsuccessfully and consecutively, before getting locked out.
- Default: 3
- Range: 1-1000
- Read-write
- Read Output Format: ["3"], integer value in quotes

3.171 Locked Timeout

- Command: locked_timeout
- Parameter: ["value"] – Locked timeout in seconds.
- This API allows the Admin to set how long in seconds a user will be locked out of the GUI after "max_auths" unsuccessful login attempts.
- Default: 60
- Range: 30-60000
- Read-write
- Read Output Format: ["60"], integer value in quotes

3.172 Idle Timeout

- Command: idle_timeout
- Parameter: ["value"] – Inactivity timeout in seconds.
- This API allows the Admin to set the inactivity timeout for the serial console. After this time, the serial session will automatically log out.
- Default: 300
- Range: 30-60000
- Read-write
- Read Output Format: ["300"], integer value in quotes

3.173 GUI Idle Timeout

- Command: gui_idle_timeout

- Parameter: ["value"] – Inactivity timeout in minutes.
- This API allows the Admin to set the inactivity timeout for the GUI. After this time, the GUI will automatically log out.
- Default: 10
- Range: 1-1000
- Read-write
- Read Output Format: ["10"], integer value in quotes

3.174 Create User

- Command: create_user
- Parameter: ["username", "role", "password"] – (Optional) "username" is the real identity of the user, "role" is the access level which can be one of "admin", "basic", or "advanced", and "password" is the plaintext password.
- This API allows the Admin to create a new user, assign their password and access level.
- If no parameters are passed, the API will return a list of existing usernames.
- Note: if a radio boots up and finds no record of any users, it will automatically add 3 users with the default password "HelloWorld":
 - "admin" with access level "admin"
 - "basic" with access level "basic"
 - "advanced" with access level "advanced"
- This will ensure compatibility with older firmware < v4.0.3.0.
- Default: N/A
- Read-write
- Read Output Format: ["admin","basic","advanced"], list of comma-separated string user names

3.175 Change Password

- Command: change_password
- Parameter: ["username", "old password", "new password"] – "username" is the username whose password will be changed, "old password" is the old password, "new password" is the new password.
- This API allows the Admin to change the password of an existing user. The old password must be given to allow the radio to enforce password complexity requirements (if enabled).
- If the Admin does not have the user's old password, they may delete and then create the user again.
- Default: N/A

- Write Only

3.176 Delete User

- Command: delete_user
- Parameter: ["username"] – Username to be deleted.
- This API allows the Admin to delete a username. Only constraint is that an Admin cannot delete their own username. This ensures that there is always at least 1 admin user in the system.
- Default: N/A
- Write-Only.

3.177 Log Out User

- Command: logout
- Parameter: ["username"] – Username to be logged out.
- This API allows the Admin to log out a username. If the Admin user passes their own username, it will immediately end their session. If the Admin user passes any other username, all their existing sessions will be immediately ended.
- Default: N/A
- Write-Only.

3.178 SSH Service

- Command: ssh_service_disable
- Parameter: ["value"] – Disable (1) or Enable (0)
- This API allows the Admin to completely shut down the SSH service. For production systems, this is the recommended configuration, to avoid vulnerabilities.
- SSH is not required for operation, only for debugging.
- Default: 0
- Write-Only.

3.179 Routing Beacon Air Time Cap

- Command: routing_beacon_max_air_pct
- Parameter: ["value"] – Percent (0-100)

- This API allows the user to set the percent of air-time that will be used for routing overhead.
- If set to 0, the radio will switch to using the “routing_beacon_period”.
- If set to a non-zero number ≤ 100 , the radio will automatically modulate the beacon sending rate so that the air-time in its 2-hop neighborhood stays below that number.
- If set to non-zero, the radio will never send more often than:
 - 1 beacon every 100ms for 20MHz bandwidth.
 - 1 beacon every 200ms for 10 MHz bandwidth.
 - 1 beacon every 400ms for 5 MHz bandwidth.
 - 1 beacon every 800ms for 2.5 MHz bandwidth.
 - 1 beacon every 1600ms for 1.25 MHz bandwidth.
- This setting is only applicable when the “routing_proto” is set to “path_vector” (Large Network Routing).
- Default: 0
- Read-Write.
- Read Output Format: [“0”], integer value in quotes

3.180 Virtual IPv6 Address

- Command: virtual_ipv6_address
- Parameter: [“vir_addr”], where vir_addr is the virtual IP address
- Description: This command will set the virtual IP address to be “vir_addr”. If no parameter is given, it will show the current virtual IP address
- Default: 2001:db8:1::1
- Read Output Format: [“2001:db8:1::1”], string ip value in quotes

3.181 Virtual IPv6 Prefix

- Command: virtual_ipv6_prefix
- Parameter: [“prefix”], where prefix is the prefix length
- Description: This command will set the prefix length. If no parameter is given, it will return the current value.
- Range: 0-128
- Default: 64
- Read Output Format: [“64”], integer value in quotes

3.182 Virtual IPv6 Gateway

- Command: virtual_ipv6_gateway
- Parameter: [“vir_gw”], where vir_gw is the virtual IP gateway
- Description: This command will set the virtual IP gateway to be vir_gw. If no parameter is given, it will show the current virtual IP gateway.
- Default: ::
- Read Output Format: [“::”], string ip value in quotes

3.183 Virtual IPv6 Disable

- Command: `virtual_ipv6_disable`
- Parameter: `["vir_ip_disable"]`, where `vir_ip_disable` is either 1 or 0
- Description: This command enable/disable virtual IP. If `vir_ip_disable` is set to 1, virtual IP will be enabled. Otherwise it will be disabled. If no parameter is given, it will return the current virtual IP status.
- Default: 1
- Read Output Format: `["1"]`, integer value in quotes

3.184 Model

- Command: `model`
- Parameter: None
- Description: Returns radio model type.
- Read-only.
- Read Output Format: `["SC4400E"]`, string ip value in quotes

3.185 Admin Pass Hash

- Command: `admin_pass_hash`
- Parameter: hash of password
- Description: sets password hash for admin user
- Readable and writable.
- **Deprecated**

3.186 Advanced Pass Hash

- Command: `advanced_pass_hash`
- Parameter: hash of password
- Description: sets password hash for advanced user
- Readable and writable.
- **Deprecated**

3.187 Basic Pass Hash

- Command: `basic_pass_hash`
- Parameter: hash of password
- Description: sets password hash for basic user
- Readable and writable.
- **Deprecated**

3.188 PTT Active 2 Mcast Group

- Command: `ptt_active_2_mcast_group`

- Parameter: ["r₀,r₁...r₄_t₀,t₁...t₄_m₀,m₁...m₄"] Variables r_i, t_i, m_i are numbers 0-15. They represent indexes to the ptt multicast groups. r_i stands for listen, t_i stands for talk, and m_i stands for monitor.
- Default: ____
- Description: Controls the second PTT unit in a dual PTT accessory. This API controls the "Inactive," "Active," "Monitor" dropdown on the PTT GUI.
Active - To make group 0 and 1 active: "0,1_0,1". (Note there must always be at least 1 active group).
Monitor - To make group 0 and 1 monitor and 2 active: "0,1,2_2_0,1". You must set r_i and m_i for monitor.
Inactive - Simply remove required index from the string. In the monitor example, 3 and 4 are inactive.
- Read Output Format: ["0_0"], list of integer values separated by underscore in quotes

3.189 PTT Vox

- Command: ptt_vox
- Parameter: enable = 1, disable = 1
- Description: Enables Voice Activated Transmission
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.190 PTT Collision Detection

- Command: ptt_collision_detect
- Parameter: enable = 1, disable = 1
- Description: Enables PTT collision detection. Beeps when attempting to talk on PTT but there is an incoming transmission.
- Default: 1
- Read Output Format: ["1"], integer value in quotes

3.191 PTT GPIO Pin Mode

- Command: ptt_gpio_pin_mode
- Parameter: 0 = ptt mode, 1 = dual ptt mode, 2 = cos mode
- Description: Selects between ptt mode, dual ptt mode, and COS mode
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.192 Local Broadcast Input Rate

- Command: mcast_xor_input_rate
- Parameter: None

- Description: Reads broadcast bps input rate
- Read only
- Read Output Format: ["10"], integer value in quotes

3.193 Local Broadcast Output Rate

- Command: mcast_xor_output_rate
- Parameter: None
- Description: Reads broadcast bps output rate
- Read only
- Read Output Format: ["11"], integer value in quotes

3.194 Local Broadcast LDPC Puncture Rate

- Command: mcast_ldpc_puncture_rate
- Parameter: 0-5
- Description: Sets the LDPC puncture rate. Defines how many parity packets to send per local broadcast batch. 0 = 200%, 1 = 100%, 2 = 50%, 3 = 33.3, 4 = 20%, 5 = 0%. For example, if set to 0, it will send 200% (2x) the number of parity packets per data packet.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.195 Local Broadcast Auto MCS Lower Bound

- Command: mcast_xor_auto_mcs_lower_bound
- Parameter: 0-4
- Description: When mcast_xor_mcs is set to 255, this will set the minimum MCS that can be selected by the auto MCS algorithm.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.196 Default Multicast Algorithm

- Command: default_mcast_algo
- Parameter: 0-3
- Description: Sets the default multicast algorithm. 0 - Legacy Multicast - transmits unicast to every destination. 1 - Broadcast - broadcast packet to all nodes. 2 - MANET Multicast - transmit multicast with additional parity packets (XOR or LDPC). 3 - Flooding Multicast - transmit multicast with packet compression.
- Default: 0
- Read Output Format: ["0"], integer value in quotes

3.197 Legacy Multicast Group List

- Command: legacy_mcast_group_list

- Parameter: ip list
- Description: Multicast addresses in this list will always use legacy multicast. Multicast addresses should only be separated by a single comma.
- Example Parameter: "231.211.175.189,238.58.254.24"
- Default: ""
- Read Output Format: ["231.211.175.189,238.58.254.24"], comma-separated ip list in quotes

3.198 Local Broadcast DSCP List Mapping

- Command: mcast_xor_dscp_list
- Parameter: DSCP list. DSCP 0-63
- Description: DSCP values in this list will always use Local Broadcast (MANET) multicast. DSCP values should only be separated by a single comma.
- Example Parameter: "40,48"
- Default: ""
- Read Output Format: ["40,48"], comma-separated dscp list in quotes

3.199 Legacy Multicast DSCP List Mapping

- Command: mcast_legacy_dscp_list
- Parameter: DSCP list. DSCP 0-63
- Description: DSCP values in this list will always use Legacy Multicast. DSCP values should only be separated by a single comma.
- Example Parameter: "40,48"
- Default: ""
- Read Output Format: ["40,48"], comma-separated dscp list in quotes

3.200 High Priority MCS

- Command: mcs_hp
- Parameter: ["mcs"], where mcs is an integer.
- Description: Sets/reads the MIMO modulation and coding rate. Setting it to 255 will enable rate adaptation.
- Range: see Table 5 in Appendix for the supported MCS. 0-15 or 255.
- Default: 255
- Read Output Format: ["255"], integer value in quotes

3.201 Critical Priority MCS

- Command: mcs_hp2
- Parameter: ["mcs"], where mcs is an integer.
- Description: Sets/reads the MIMO modulation and coding rate. Setting it to 255 will enable rate adaptation.
- Range: see Table 5 in Appendix for the supported MCS. 0-15 or 255.
- Default: 255
- Read Output Format: ["255"], integer value in quotes

3.202 Infrastructure Mesh Failover Mode

- Command: mesh_failover_mode
- Parameter: ["mode"] where mode is 0 (External i.e. cradlepoint router), 1 (N2N, VPN), or 2 (Legacy)
- Description: Sets behavior when SNR falls below mobile_lte_failover_snr_thresh. When set to 0, radio prevents ping packet forwarding, and relies on the user setting up an external WAN. When set to 1, enable the vpn client as the failover backup. When set to 2, it doesn't take any action when the threshold is crossed.
- Default: 2
- Read Output Format: ["2"], integer value in quotes

3.203 Infrastructure Mesh Failover SNR Threshold

- Command: mobile_lte_failover_snr_thresh
- Parameter: ["snr"]
- Description: When SNR to the infrastructure node falls below this threshold, it triggers the failover link.
- Range: -100 to 100
- Default: -5
- Read Output Format: ["-5"], integer value in quotes

3.204 Voltage Reporting Address

- Command: voltage_reporting_address
- Parameter: ["IP", "Port"], where IP is an IP address in the format of XXX.XXX.XXX.XXX, port is an integer.
- Description: Sets/reads the voltage reporting IP address and port.
- Range: XXX in the IP address should be between 0 and 255, port number cannot exceed 65535, it is suggested to use a port number around 2000.
- Default: "10.1.1.3", "30000"
- Read Output Format: ["10.1.1.3", "30000"], string ip and integer port values in quotes

3.205 Voltage Reporting Threshold

- Command: voltage_reporting_min_threshold and voltage_reporting_max_threshold
- Parameter: ["voltage"], where voltage is an integer.
- Description: Sets/reads the Min/Max voltage threshold. When the voltage exceeds the "voltage_reporting_max_threshold" it is considered to be in an over voltage state. When the radio voltage is below voltage_reporting_min_threshold, it is considered to be in an under voltage state.
- Unit: millivolts.
- Default: min 10000, max 34000

- Range: 0-100000
- Resolution: 1
- Read Output Format: ["10000"], integer value in quotes

3.206 Voltage Reporting Period

- Command: voltage_reporting_period
- Parameter: ["period"], where period is an integer.
- Description: Sets/reads the voltage reporting time period. Reports are sent to voltage_reporting_address every "period" seconds.
- Unit: second.
- Range: 1-600 (upper limit of 32bit integer)
- Resolution: 1
- Default: 5 seconds.
- Read Output Format: ["5"], integer value in quotes

3.207 Voltage Reporting Mode

- Command: voltage_reporting_mode
- Parameter: ["mode"], where mode is either 0, 1, or 2.
- Description: Sets/reads the voltage report mode. 0 disables the report, 1 enables the report only when the radio exceeds the max voltage or is below the min voltage, 2 sends the report periodically every voltage_reporting_period seconds.
- Default: 0 (disable)
- Read Output Format: ["0"], integer value in quotes

3.208 Link Statistics – MCS

- Command: link_stats_mcs
- Parameter: ["nodeid"] – the nodeid of the neighbor node
- Returns the MCS for the specific link. This is the same statistics visible in the GUI section: "Network Management" -> "Network Topology" -> "Link Statistics" (click on the individual radio links to view the "Link Statistics" popup).
- Read-only
- Read Output Format: ["12"], integer value in quotes

3.209 Add Settings Profile

- Command: add_profile
- Parameter: {"profile_name": "name", "profile_file": "file"}
- Description: Adds the provided profile "profile_file" onto the radio. Saved as "profile_name".
- Write-only

3.210 PTT Mcast Name

- Command: `ptt_mcast_name`
- Parameter: ["index", "name"] where index is 0-15 and represents the Multicast Groups 1-16 on the GUI. Name is the given ptt group name.
- Description: It's a text description for each ptt group. To read the value of this command, provide the index. For example, "`ptt_mcast_name 0`".
- Default: ["0", "PTT Group 1"]
- Read Output Format: ["PTT Group 1"], string group name value in quotes

3.211 Encryption RF Broadcast Authorization Key

- Command: `enc_rf_bcast_auth_key`
- Parameter: ["key"] key length must be 64 hexadecimal characters.
- Description: This is the key used for broadcast packets when `enc_profile` is 27 (AES GCM ECDH-KAS), when `enc_profile` is 26, it is the key used for all packets (unicast and broadcast).
- Default:
"2345678123456781234567812345678123456781234567812345678123456781"
- Example:
"7ec97ae69117720d0ad9213535d6b71a59fcbd27bca997019fa1c9e33f09ddca"
- Read Output Format: ["7ec97ae691..."], string key value in quotes

3.212 Encryption HMAC Key

- Command: enc_hmac_key
- Parameter: ["key"] key length must be 64 hexadecimal characters.
- Description: This is the HMAC-SHA-256 key used to sign the json request.
If correctly signed, the api will be executed with admin privileges
- Default:
"3456781234567812345678123456781234567812345678123456781234567812"
- Example:
"a7f8f111dbae395891dcc066c27f66c87cc3f4bfac97b1e17dd528eb3b8c1c74"
- Read Output Format: ["7ec97ae691..."], string key value in quotes

3.213 FIPS Ready

- Command: `fips_ready`
- Description: Returns `[""]` if the radio has successfully entered FIPS approved mode. Else it will return a list of APIs that need to be updated to transition to the FIPS approved state, e.g. set new encryption keys, new https certificate, and all passwords (admin, advanced, and basic) must be changed from the default with password complexity enabled and minimum password length of 8 characters. If `"fips_mode"` is set to 0, this will still return `[""]`.
- This API is read-only.

- Read Output Format: ["Default value APIs:admin,enc_rf_auth_key,"], comma separated list of APIs still containing default values in quotes

3.214 Deferred Execution API

- Command: deferred_execution_api
- Parameter: {"version":"version_num", "sleep":"sleep_time", "api_list":api_array}. "sleep_time" time in seconds to wait before executing the APIs. api_array is the list of APIs to execute.
- Description: Executes the list of APIs provided after sleep_time expires.
- Example:

```
{"version":"2.0","sleep":"0","api_list":[{"jsonrpc":"2.0","method":"max_link_distance","params":["4444"],"id":"1"}, {"jsonrpc":"2.0","method":"mcs","params":["4"],"id":"2"}]}
```
- This API is write-only.

3.215 Routing Tree

- Command: routing_tree
- This API is read-only
- Description: Returns list of all radios in the network.
- Read Output Format: [3058,2920,2977,2994,22103], list of integers

3.216 QoS Class CW Min

- Command: qos_class_cwmin
- Parameters: ["qos_class", "cw_min"]. "qos_class" is 0 (normal), 1 (high), or 2 (critical). "cw_min" is an integer in the range 3 to 10
- Description: Sets the QoS contention window minimum for a particular class.
- Example: ["0", "3"]
- Read Output Format: ["3"], integer enclosed in quotes

3.217 QoS Class CW Max

- Command: qos_class_cwmax
- Parameters: ["qos_class", "cw_max"]. "qos_class" is 0 (normal), 1 (high), or 2 (critical). "cw_max" is an integer in the range 3 to 10
- Description: Sets the QoS contention window maximum for a particular class.
- Example: ["0", "4"]
- Read Output Format: ["4"], integer enclosed in quotes

3.218 Broadcast Discovery Configuration

- Command: `bcast_discovery_config`
- Parameter: `[enable_state, interval_ms, dest_ip, source_port, dest_port, dest_mac]`
- Description: Configures the broadcast discovery daemon.
- Range: `enable_state`: `enable` = "1", `disable` = "0". `interval_ms` = 1000-600000. `Dest_ip` = "0.0.0.0" to "255.255.255.255". `Source_port` = 0-65535. `Dest_port` = 0-65535. `Dest_mac` = "00:00:00:00:00:00" to "FF:FF:FF:FF:FF:FF".
- Default: `"[\"0\", \"1000\", \"255.255.255.255\", \"20000\", \"20000\", \"FF:FF:FF:FF:FF:FF\"]"`
- Example: `"[\"1\", \"3000\", \"255.255.255.255\", \"20000\", \"20000\", \"FF:FF:FF:FF:FF:FF\"]"`

3.219 MAN-SA Is Root

- Command: `is_root`
- Description: Sets state of `is_root`. Value "1" generates a new `nw_info` file for MAN-SA.
- Parameter: `["state"]`, expected values [0,1]
- Example: `["1"]`
- Read/Write

3.220 BCAST Update Status

- Command: `bcast_update_enable`
- Description: Checks for on-going bcast. If enabled, disables network management bcast updates.
- Parameter: `["state"]`, expected values [0,1]
- Example: `["1"]`
- Value of "1" means bcast is currently in progress.
- Read/Write
- May no longer be used since converting Network-Wide Setup to using `bcast_enc.py`

3.221 Silvus Line Layer

- Command: `line_layer`
- Description: Checks if the Map Overlay is using the Silvus Line Layer.
- Parameter: `["state"]`, expected values [0,1]
- Example: `["1"]`
- Value of "1" means that the Map Overlay is currently using the Silvus Line Layer

- Read/Write

3.222 Default Latitude and Longitude

- Command: default_lat_lon
- Description: Sets defaultLat and defaultLon in flash.
- Parameters: ["lat","lon"], expects 2 parameters to set.
- Example: ["34.05","-118.44"]
- Read/Write

4 Streaming Responses

Some users may be interested in streaming specific information from the radio e.g. rssi, noise floor, temperature, etc. After enabling the response they need using the above commands, the radio will transmit the desired information in the form of UDP packets to a specific IP address and port. The format of each report message will be in the type-length-value format as shown below:

TYPE LENGTH VALUE TYPE LENGTH VALUE ...

- TYPE and LENGTH will be 16-bit unsigned integers in network-endian format.
- TYPE indicates the kind of information being transmitted. Pre-defined types are listed later in this document.
- LENGTH indicates the length of the VALUE field in bytes, including the terminating null byte.
- VALUE will be ASCII-encoded text terminated with a null byte ('\0').
- A single report will comprise of a set of type-length-value fields beginning with a "begin" report type. It will have a type which is specific to the type of report being generated, length of 1 byte and a value of an empty string (""). Note the empty string is still null terminated.
- Each report will end with an end of report which has type 1 (type = end of report, length = 1, value = "").
- The empty string listed above has a NULL character and has length 1. Any length number in the streaming report includes the NULL character
- A UDP packet may contain more than one report.
- The UDP packets have a maximum size of 1400 bytes.

5 RSSI and Noise Floor Reporting

The type/length/value for RSSI and noise floor reporting are listed in the following table:

Table 1. RSSI Reporting Format

Report Type	Data Type	Information
5009	Empty string ""	Begin of RSSI report
5010	Float	Revision number for RSSI report
5000	Integer	Raw signal power of first antenna (for unit see note below)
5001	Integer	Raw signal power of second antenna (for unit see note below)
5002	Integer	Raw signal power of third antenna (for unit see note below)
5003	Integer	Raw signal power of fourth antenna (for unit see note below)
5004	Integer	Raw noise power (for unit see note below)
5005	32-bit integer	Sync signal power (from digital domain, see note below).
5006	32-bit integer	Sync noise power (from digital domain, see note below).
5007	16-bit integer	Node ID of the radio.
5008	32-bit integer	Report sequence number, increments for every report, resets after 9999.
5011	String	Node IP
5012	String	Virtual IP
1	Empty string ""	End of report.

Note:

The raw signal/noise power are in half dB steps if PHY version is <1.6, and are in dB steps if PHY version is >= 1.6. PHY version may be checked with the API "fpga_build_info". Additionally, any release >= 3.12.4.13 will use the 1dB steps format.

The sync noise and power (types 5005, 5006) are special values obtained after packet processing in the digital domain. They cannot be directly compared to the raw signal and noise values. To obtain an SNR from these values the user needs to run the below formula on these values:

X = sync signal power;

Y = sync noise power;

$$Z = (Y - X) / 51$$

$$\text{SNR}_{\text{mw}} = (X - 12 * Z) / (64 * Z)$$

$$\text{SNR}_{\text{db}} = 10 * \log(\text{SNR}_{\text{mw}}) / \log(10)$$

SNR_db is the SNR in dB and it is averaged across all antennae.

The SNR obtained above is more accurate when the real SNR goes below 10 dB. Above 10 dB, the SNR obtained from the raw signal and noise values are more accurate. The SNR obtained via sync signal and noise power will be severely inaccurate above 10dB.

Below is an example of the RSSI report:

Table 2. Sample RSSI Report

Report Type	Length	Information
5009	1	""
5010	4	"1.0"
5008	5	"2333"
5000	5	"-43"
5001	5	"-31"
5002	5	"-28"
5003	5	"-66"
5004	5	"-190"
5005	8	"8604568"
5006	8	"8861322"
5007	5	"1025"
1	1	""

The corresponding raw UDP dump in hexadecimal format is attached below. For the purpose of easier reading, each byte is separated by a space, and each item is separated by a new line. The real streaming report is continuous without any spaces or newlines and is currently 109 bytes long.

```

13 fffff91 0 1 0
13 fffff92 0 4 31 2e 30 0
13 fffff90 0 5 32 33 33 33 0
13 fffff88 0 5 20 2d 34 33 0
13 fffff89 0 5 20 2d 33 31 0
13 fffff8a 0 5 20 2d 32 38 0
13 fffff8b 0 5 20 2d 36 36 0
13 fffff8c 0 5 2d 31 39 30 0
13 fffff8d 0 a 20 20 38 36 30 34 35 36 38 0
13 fffff8e 0 a 20 20 38 38 36 31 33 32 32 0
13 fffff8f 0 5 31 30 32 35 0
0 1 0 1 0

```

Note that RSSI reports are only generated if there's traffic between the radios. Not enough traffic with short RSSI report period chosen will result in the user receiving RSSI reports with all values set to 999, which is designed to let the user know the radio link is still alive.

6 Temperature Reporting

The type/length/value for temperature reporting are listed in the following table:

Table 3. Temperature Reporting Format

Report Type	Data Type	Data
8	Empty string ""	Begin of temperature report.
9	Float	Revision number for temperature report.
2	Integer	Current Temperature on the radio.
5	Integer	Node ID
6	String	Node IP
7	String	Virtual IP
3	Integer	Maximum Temperature reached on the radio after last booting.
4	Integer	Overheat Count: number of times the radio temperature has exceeded temp_reporting_max_threshold.
1	Empty string ""	End of report

7 Spectrum Scanning

The type/length/value for spectrum scan reporting are listed in the following table:

Table 4. Spectrum Scan Reporting Format

Report Type	Data Type	Information
6000	Empty string ""	Begin of Spectrum Scan report
6001	Float	Revision number for Spectrum Scan Report
6002	Float	Frequency in MHz
6003	16-bit Integer	FFT size
6007	16-bit Integer	Antenna index, 0~3
6004	Comma separated float numbers	Averaged spectrum, an array of FFT size.
6005	Comma separated float numbers	Peak spectrum, an array of FFT size.

6006	Comma separated integer numbers	Threshold crossover count, an array of FFT size.
Repeat field 6007, 6004, 6005 and 6006 if needed.		
1	Empty string ""	End of report.

The spectrum scan report basically contains 3 arrays for each frequency scanned. One frequency can have multiple captures, with fast Fourier transform performed on every one of them. The first array is called “averaged spectrum”, which is every FFT result captured for that frequency averaged together. The second array is called “peak spectrum”, it records the peak value of each spectrum over all captures corresponding to that frequency. The last array is the threshold(specified by user) crossover count for each spectrum over all the capture’s FFT. These 3 arrays all have the same size as the chosen FFT size. The array values are sent one by one with a comma in between, so there will be one long string containing several comma separated numbers. Note the unit of FFT output is converted to dBm/Hz and reported.

8 Rebrand

The radio supports rebranding of the logo and user manual. The user can access the rebrand interface by entering the following address in the browser:

<http://<IP>/rebrand.sh>

The interface allows the user to upload custom logo and manual. The logo should be in PNG format and named as “silvus_logo_final.png”, the manual should be in PDF format and named as “user_manual.pdf”. The upload file size cannot be larger than 2.5MB. For the reference, 1413x557 is the original logo dimension.

After uploading custom files, the changes will take effect immediately. To remove the rebranding, simply click the “Remove” button on the interface, and the rebranding will be removed after a reboot.

9 Password Protected API Access

API access is password protected using the same login password mechanism that is used to protect the web interface. (Note: login password protection feature is only available on SC4200, SC4400 and SC3822.). Currently only the parameters on the "Security" tab are password protected when using the API.

Follow the steps below:

1. Setup the login password for user "admin" using the web interface.
2. Use a cookie enabled http client e.g. curl to issue API requests.

3. Authenticate with the radio using the "admin" username and the password set in Step 1. If the credentials are correct the radio web server will return a "cookie" that is used to authenticate future requests. The password used in below example is "admin".

```
curl -skL "http://172.20.4.34/login.sh?username=admin&password=admin&Submit=1" -c cookie.jar
```

4. The returned cookie (cookie.jar) is time sensitive and will expire in 10 minutes.

5. Every API request thereafter returns a refreshed cookie that will last another 10 minutes.

```
curl -skL -X POST -d '{"jsonrpc":"2.0", "method":"enc_disable", "id":"0"}' "http://172.20.4.34/streamscape_api" -c cookie.jar -b cookie.jar
```

Appendix

Table 5. Supported Modulation and Coding Index (MCS)

MCS	Number of Spatial Streams	Constellation	FEC Rate	PHY Throughput for 20MHz BW (Mbps)
0	1	BPSK	1/2	6.5
1	1	QPSK	1/2	13
2	1	QPSK	3/4	19.5
3	1	16QAM	1/2	26
4	1	16QAM	3/4	39
5	1	64QAM	2/3	52
6	1	64QAM	3/4	58.5
8	2	BPSK	1/2	13
9	2	QPSK	1/2	26
10	2	QPSK	3/4	39
11	2	16QAM	1/2	52
12	2	16QAM	3/4	78
13	2	64QAM	2/3	104
14	2	64QAM	3/4	117
255	Auto	Auto	Auto	

The following MCSs are included in the auto rate adaptation: 0-6, 8-15.

Note the actual usable MCS depends on channel environment, signal to noise ratio, antenna configuration, and other factors.

10 Local Access API

Silvus radios listen on port 50000 for local operations that cannot be accessed over the mesh. Only API calls originating from local network interfaces (Ethernet, USB) will be allowed through. These APIs **do not need authentication**.

A server listens on JSON/RPC port 50000 and implements the following three services:

1. "zeroize" : This is the same as the zeroize api call. It will clear all sensitive information like passwords, encryption keys, etc.
2. "radio_reset": This will reboot the radio after three seconds.
3. "factory_reset": This will reset all radio settings to factory default. The default settings will only take effect after a reboot, so right after this call, the radio needs to be rebooted manually or via API.

Here is an example using the python jsonrpclib library to completely reset all passwords and settings:

```
import jsonrpclib
p = jsonrpclib.Server("http://172.20.68.153:50000")
p.factory_reset()
p.zeroize()
p.radio_reset()
```

11 Voltage Reporting

The type/length/value for voltage reporting are listed in the following table:

Table 6. Voltage Reporting Format

Report Type	Data Type	Data
4001	Empty string ""	Start of Report
4003	Float	Revision number for report.
4004	Float	Current Voltage on the radio.
4005	Float	Minimum voltage seen on the radio since last boot up.
4006	Float	Maximum voltage seen on the radio since last boot up.
4007	Integer	Undervoltage Count: number of times the radio temperature has exceeded <code>voltage_reporting_max_threshold</code> .
4008	Integer	Overvoltage Count: number of times the radio temperature has exceeded <code>voltage_reporting_min_threshold</code> .
1	Empty string ""	End of report

12 API User Privilege

This section lists the API restrictions by user type. The Admin section lists the APIs only available to the Admin user. Similarly, the Advanced section lists the APIs only available to the Advanced user. The 'No Restriction APIs' are available to all users.

12.1 Admin User Only APIs

```
add_ssh_key
admin_pass
advanced_pass
basic_pass
black_list
change_password
console_notice_enable
create_user
```


delete_user
deferred_execution_api
disable_concurrent_sessions
enable_api_logs
enable_password_complexity
enc_disable
enc_disable_confirm
enc_hmac_key
enc_key
enc_key_len
enc_key_volatile_disable
enc_profile
enc_rf_auth_key
enc_rf_bcast_auth_key
fips_mode
gui_idle_timeout
host_key
idle_timeout
led_brightness
led_disable
list_ssh_keys
locked_timeout
max_auths
min_password_change
min_password_length
notice_of_use
radio_reset
remove_ssh_key
rssi_report_address
rssi_report_enable
rssi_report_period
send_log_to
ssh_service_disable
stdsa_camp_freq_fixed_index
stdsa_camp_freq_selection_mode
temp_reporting_address
temp_reporting_max_threshold
temp_reporting_min_threshold
temp_reporting_mode
temp_reporting_period
tls_cert
voltage_reporting_address
voltage_reporting_max_threshold
voltage_reporting_min_threshold
voltage_reporting_mode
voltage_reporting_period

wb_list_type
white_list
zeroize

12.2 Advanced User Only APIs

mps_disable
mps_profile
mps_pos
mps_freq

12.3 All Users APIs

add_profile
admin_pass_hash
advanced_pass_hash
aggr_thresh
basic_pass_hash
battery_percent1
beamform_enable
build_information
build_tag
burst_time
bw
current_tof
custom_mcast_filter_enable
cw_start
cw_stop
default_mcast_algo
enable_max_power
fips_ready
fpga_build_info
freq
freq_bw
generate_host_key
generate_tls_cert
get_profile
gps_coordinates
gps_mode
gps_time
igmp_snoop_enable
igmp_snoop_mesh_forwarding_mode
input_voltage_monitoring
la_mode
legacy_mcast_group_list
link_stats_mcs

link_stats_total_air_time
link_stats_total_loss
link_stats_total_rate
link_stats_var_gi_cyc_pre_len
link_stats_var_gi_mode
link_throughput
list_of_profiles
local_address
local_stats_fw_bc_mc
local_stats_fw_uc
local_stats_input_bc_mc
local_stats_input_dropped
local_stats_input_uc
local_stats_interference
logout
max_link_distance
max_speed
mcast_config
mcast_ldpc_puncture_rate
mcast_legacy_dscp_list
mcast_mhop_mode
mcast_one_receiver_enable
mcast_xor_aggr_thresh
mcast_xor_auto_mcs_lower_bound
mcast_xor_block_size
mcast_xor_disable
mcast_xor_dscp_list
mcast_xor_group_list
mcast_xor_input_rate
mcast_xor_mcs
mcast_xor_output_rate
mcast_xor_rate_mode
mcast_xor_target_latency
mcs
mcs_hp
mcs_hp2
mesh_failover_mode
mesh_node_type
mobile_lte_failover_snr_thresh
mobile_mesh_scan_list
mobile_mesh_scan_result
model
mps_supported
nbr_mcs
nbr_mcs_rx
nbr_rssi

network_status
node_label
node_labels
nodeid
noise_estimate_enable
noise_level
num_retx
num_retx_hp
num_retx_hp2
num_rts_retx
nw_name
opus_rate
pa_basic_mode_max_output_power_per_channel
pa_basic_mode_settings
pa_mode
power_dBm
print_all_settings
ptt_active_2_mcast_group
ptt_active_mcast_group
ptt_aggr_delay
ptt_audio_encoder_type
ptt_beep_volume
ptt_collision_detect
ptt_gpio_pin_mode
ptt_hq_link_notifications
ptt_level_change_link_notifications
ptt_levels_link_notifications
ptt_mcast_group
ptt_mcast_name
ptt_mic_bias
ptt_mic_type
ptt_mic_vol
ptt_no_link_notifications
ptt_notification_volume
ptt_speaker_vol
ptt_status
ptt_vox
qos_class
qos_class_reset
queue_size
radio_mode
read_current_temperature
read_power_dBm
read_power_mw
response_disable
routing_beacon_max_air_pct

routing_beacon_mcs
routing_beacon_period
routing_path
routing_proto
rts_disable
rx_ant_mask
serial_config
setenvlinsingle
spectrum_scan_start
spectrum_scan_stop
stdsa_freq_config
stdsa_recommended_frequency
stdsa_scan
supported_frequencies
supported_frequency_profiles
tx_ant_mask
tx_fifo_disable
var_gi_cyc_pre_len
var_gi_mode
version
virtual_ip_address
virtual_ip_disable
virtual_ip_gateway
virtual_ip_netmask
virtual_ipv6_address
virtual_ipv6_disable
virtual_ipv6_gateway
virtual_ipv6_prefix
vpn_address
vpn_disable
wbg_disable
weakest_link

13 FIPS Restrictions

FIPS has strict security requirements which must be fulfilled for the radio to operate in FIPS mode. The following API restrictions are enforced:

- The following user password restrictions are enforced:
- User (basic, advanced, and admin) passwords must be changed from the default HelloWorld
- Password complexity must be enabled (enable_password_complexity set to 1)
- min_password_length must be greater than or equal to 8
- The following keys/certificates must be change from their defaults:

- `enc_rf_bcast_auth_key`
- `enc_rf_auth_key`
- `enc_hmac_key`
- `tls_cert`
- Encryption must be enabled (`enc_disable` set to 0)
- Only these encryption profiles are supported: AES GCM Static 256-bit (26, 256) and AES GCM ECDH 256-bit (27, 256). Applies to `enc_profile` API.
- SSH must be disabled (`ssh_service_disable` set to 1)